UN R137 (version 4.1.0)

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0. Introduction

0.1. For the 04 series of amendments:

0.1.1. Currently, under UN Regulation No. 137 in its 03 series of amendments, to protect vehicle occupants of age 65 and older, the Thorax Compression Criterion (ThCC) for the Hybrid III American Female fifth percentile dummy (AF05) is limited to maximum 34 mm in vehicles of category M_1[1].

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.8 - https://unece.org/transport/vehicle-regulations/wp29/resolutions

0.1.2. When developing this new 04 series of amendments to UN Regulation No. 137, the Working Party on Passive Safety (GRSP) considered it desirable to enhance the protection of occupants of age 65 and older by applying the same AF05 dummy ThCC limit of 34 mm also to a certain range of vehicles of category N_1[1] which often may be used in specific countries by private consumers for commuting.

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.8 - https://unece.org/transport/vehicle-regulations/wp29/resolutions

0.1.3. Some national accident data indicate that almost all the vehicles in which occupants of age 65 and older in front were fatally or seriously injured were very small N_1 vehicles only available in a certain number of countries. This amendment therefore seeks to address this particular case.

0.2. For supplement 1 to the 04 series of amendments:

0.2.1. The Regulation is amended to account for vehicles of category X[1]. Vehicles of category Y[1] are not in the scope of this Regulation.

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.8 - https://unece.org/transport/vehicle-regulations/wp29/resolutions

0.2.2. The Regulation was originally drafted for vehicles with driver and manual driving controls. It is the intention of this amendment to keep the spirit of the regulation and to extend its application to vehicles without driver and without manual driving controls inside the vehicle. In the absence of a driver or manual driving controls in the vehicle, provisions related to them shall not be taken into account if not already covered by this amendment.

0.2.3. In case of vehicles equipped with an Automated Driving System (ADS)[1] other than vehicles of category X, in the manual driving mode no special provisions or exemptions apply. In a mode where an ADS feature is active the relevant ADS requirements apply.

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.8 - https://unece.org/transport/vehicle-regulations/wp29/resolutions

1. Scope

1.1. This Regulation applies to vehicles of category M_1 with a maximum permissible mass not exceeding 3,500 kg and to vehicles of category N_1.

1.2. This regulation does not apply to vehicles of category Y.

2. Definitions

For the purpose of this Regulation:

2.1. "*Protective system*" means interior fittings and devices intended to restrain the occupants and contribute towards ensuring compliance with the requirements set out in paragraph 5. below.

2.2. "*Type of protective system*" means a category of protective devices which do not differ in such essential respects as their:

(a) Technology;

(b) Geometry;

(c) Constituent materials.

2.3. "*Vehicle width*" means the distance between two planes parallel to the longitudinal median plane (of the vehicle) and touching the vehicle on either side of the said plane but excluding the external devices for indirect vision, side marker lamps, tyre pressure indicators, direction indicator lamps, position lamps, sensors for Advanced Driver Assistance Systems[1], sensors for Automated Driving Systems, flexible mud-guards and the deflected part of the tyre side-walls immediately above the point of contact with the ground.

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.8 - https://unece.org/transport/vehicle-regulations/wp29/resolutions

2.4. "*Vehicle type"* means a category of power-driven vehicles which do not differ in such essential respects, in so far as they have an adverse effect on the result of the impact test prescribed in this Regulation, as:

(a) The length and width of the vehicle;

(b) The structure, dimensions, lines and materials of the part of the vehicle forward of the transverse plane through the "R" point of the driver's seat. For vehicles of category X, the "R" point of the most forward outboard passenger seat shall be taken instead of the R-point of the driver's seat;

(c) The lines and inside dimensions of the passenger compartment and the type of protective system;

(d) The siting (front, rear or centre) and the orientation (transversal or longitudinal) of the engine;

(e) The unladen mass;

(f) The optional arrangements or fittings provided by the manufacturer;

(g) The locations of the REESS.

(h) The basic configuration and main characteristics of the compressed hydrogen storage system.

2.5. Passenger compartment

2.5.1. "*Passenger compartment with regard to occupant protection*" means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, outside glazing and front bulkhead and the plane of the rear compartment bulkhead or the plane of the rear-seat back support;

2.5.2. "*Passenger compartment for electric safety* and/or hydrogen safety *assessment"* means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, outside glazing, front bulkhead and rear bulkhead, or back door, as well as by the electrical protection barriers and enclosures provided for protecting the occupants from direct contact with high voltage live parts.

2.6. "*R point*" means a reference point defined for each seat by the manufacturer in relation to the vehicle's structure, as indicated in Annex 6.

2.7. "*H point*" means a reference point determined for each seat by the testing service responsible for approval, in accordance with the procedure described in Annex 6.

2.8. "*Unladen kerb mass*" means the mass of the vehicle in running order, unoccupied and unladen but complete with fuel, coolant, lubricant, tools and a spare wheel (if these are provided as standard equipment by the vehicle manufacturer).

2.9. "*Airbag*" means a device installed to supplement safety belts and restraint systems in power-driven vehicles, i.e. systems which, in the event of a severe impact affecting the vehicle, automatically deploy a flexible structure intended to limit, by compression of the gas contained within it, the gravity of the contacts of one or more parts of the body of an occupant of the vehicle with the interior of the passenger compartment.

2.10. "*Passenger airbag*" means an airbag assembly intended to protect occupant(s) in seats other than the driver's seat in the event of a frontal collision.

2.11. "*High voltage*" means the classification of an electric component or circuit, if its working voltage is > 60 V and ≤ 1,500 V direct current (DC) or > 30 V and ≤ 1,000 V alternating current (AC) root - mean - square (rms).

2.12. "*Rechargeable Electrical Energy Storage System (REESS)*" means the rechargeable energy storage system that provides electric energy for electrical propulsion.

A battery whose primary use is to supply power for starting the engine and/or lighting and/or other vehicle auxiliaries' systems is not considered as a REESS.

The REESS may include the necessary systems for physical support, thermal management, electronic controls and casing.

2.13. "*Electrical protection barrier*" means the part providing protection against any direct contact to the high voltage live parts.

2.14. "*Electric power train*" means the electrical circuit which includes the traction motor(s), and may also include the REESS, the electrical energy conversion system, the electronic converters, the associated wiring harness and connectors, and the coupling system for charging the REESS.

2.15. "*Live parts*" means conductive part(s) intended to be electrically energized under normal operating conditions.

2.16. "*Exposed conductive part*" means the conductive part which can be touched under the provisions of the protection degree IPXXB and which is not normally energized, but which can become electrically energized under isolation failure conditions. This includes parts under a cover that can be removed without using tools.

2.17. "*Direct contact*" means the contact of persons with high voltage live parts.

2.18. "*Indirect contact*" means the contact of persons with exposed conductive parts.

2.19. "*Protection degree IPXXB*" means protection from contact with high voltage live parts provided by either an electrical protection barrier or an enclosure and tested using a Jointed Test Finger (IPXXB) as described in paragraph 4. of Annex 9.

2.20. "*Working voltage*" means the highest value of an electrical circuit voltage root-mean-square (rms), specified by the manufacturer, which may occur between any conductive parts in open circuit conditions or under normal operating conditions. If the electrical circuit is divided by galvanic isolation, the working voltage is defined for each divided circuit, respectively.

2.21. "*Coupling system for charging the rechargeable electrical energy storage system (REESS)*" means the electrical circuit used for charging the REESS from an external electrical power supply including the vehicle inlet.

2.22. "*Electrical chassis*" means a set made of conductive parts electrically linked together, whose electrical potential is taken as reference.

2.23. "*Electrical circuit*" means an assembly of connected live parts which is designed to be electrically energized in normal operation.

2.24. "*Electrical energy conversion system*" means a system (e.g. fuel cell) that generates and provides electrical energy for electrical propulsion.

2.25. "*Electronic converter*" means a device capable of controlling and/or converting electrical power for electrical propulsion.

2.26. "*Enclosure*" means the part enclosing the internal units and providing protection against any direct contact.

2.27. "*High voltage bus*" means the electrical circuit, including the coupling system for charging the REESS, that operates on a high voltage. Where electric circuits, that are galvanically connected to each other and fulfil the specific voltage condition, only the components or parts of the electric circuit that operate on high voltage are classified as a high voltage bus.

2.28. "*Solid insulator*" means the insulating coating of wiring harnesses, provided in order to cover and prevent the high voltage live parts from any direct contact.

2.29. "*Automatic disconnect*" means a device that when triggered, galvanically separates the electrical energy sources from the rest of the high voltage circuit of the electric power train.

2.30. "*Open type traction battery*" means a type of battery requiring filling with liquid and generating hydrogen gas that is released to the atmosphere.

2.31. "Automatically activated door locking system" means a system that locks the doors automatically at a pre-set speed or under any other condition as defined by the manufacturer.

2.32. "*Displacement system*" means a device by which the seat or one of its parts can be displaced and/or rotated, without a fixed intermediate position, to permit easy access of occupants to and from the space behind the seat concerned.

2.33. "*Aqueous electrolyte*" means an electrolyte based on water solvent for the compounds (e.g. acids, bases) providing conducting ions after its dissociation.

2.34. "*Electrolyte leakage*" means the escape of electrolyte from the REESS in the form of liquid.

2.35. "*Non-aqueous electrolyte*" means an electrolyte not based on water as the solvent.

2.36. "*Normal operating conditions*" include operating modes and conditions that can reasonably be encountered during typical operation of the vehicle including driving at legally posted speeds, parking and standing in traffic, as well as, charging using chargers that are compatible with the specific charging ports installed on the vehicle. It does not include, conditions where the vehicle is damaged, either by a crash, road debris or vandalization, subjected to fire or water submersion, or in a state where service and or maintenance is needed or being performed.

2.37. "*Specific voltage condition*" means the condition that the maximum voltage of a galvanically connected electric circuit between a DC live part and any other live part (DC or AC) is ≤ 30 V AC (rms) and ≤ 60 V DC.

Note 1: When a DC live part of such an electric circuit is connected to electrical chassis and the specific voltage condition applies, the maximum voltage between any live part and the electrical chassis is ≤ 30 V AC (rms) and ≤ 60 V DC.

Note 2: For pulsating DC voltages (alternating voltages without change of polarity) the DC threshold shall be applied.

2.38. "*State of Charge (SOC)*" means the available electrical charge in a REESS expressed as a percentage of its rated capacity.

2.39. "*Fire*" means the emission of flames from the vehicle. Sparks and arcing shall not be considered as flames.

2.40. "*Explosion*" means the sudden release of energy sufficient to cause pressure waves and/or projectiles that may cause structural and/or physical damage to the surrounding of the vehicle.

2.41. "*Compressed hydrogen storage system (CHSS)"* means a system designed to store compressed hydrogen fuel for a hydrogen-fuelled vehicle and composed of a container, container attachments (if any), and all primary closure devices required to isolate the stored hydrogen from the remainder of the fuel system and the environment.

2.42. "*Container"* (for hydrogen storage) means the pressure-bearing component on the vehicle that stores the primary volume of hydrogen fuel in a single chamber or in multiple permanently interconnected chambers.

2.43. "*Container Attachments"* mean non-pressure bearing parts attached to the container that provide additional support and/or protection to the container and that may be only temporarily removed for maintenance and/or inspection only with the use of tools.

2.44. "*Hydrogen-fuelled vehicle"* means any motor vehicle that uses compressed gaseous hydrogen as a fuel to propel the vehicle, including fuel cell and internal combustion engine vehicles. Hydrogen fuel for the vehicles is specified in ISO 14687:2019 and SAE J2719_202003.

2.45 "*Shut-off valve (for hydrogen-fuelled vehicles)"* means a valve between the container and the vehicle fuel system that must default to the "closed" position when not connected to a power source.

3. Application for approval

3.1. The application for approval of a vehicle type with regard to the protection of the occupants of the front seats in the event of a frontal collision shall be submitted by the vehicle manufacturer or by their duly accredited representative.

3.2. It shall be accompanied by the undermentioned documents in triplicate and following particulars:

3.2.1. A detailed description of the vehicle type with respect to its structure, dimensions, lines and constituent materials;

3.2.2. Photographs, and/or diagrams and drawings of the vehicle showing the vehicle type in front, side and rear elevation and design details of the forward part of the structure;

3.2.3. Particulars of the vehicle's unladen kerb mass;

3.2.4. The lines and inside dimensions of the passenger compartment;

3.2.5. A description of the interior fittings and protective systems installed in the vehicle;

3.2.6. A general description of the electrical power source type, location and the electric power train (e.g. hybrid, electric).

3.3. The applicant for approval shall be entitled to present any data and results of tests carried out which make it possible to establish that compliance with the requirements can be achieved with a sufficient degree of confidence.

3.4. A vehicle which is representative of the type to be approved shall be submitted to the Technical Service responsible for conducting the approval tests.

3.4.1. A vehicle not comprising all the components proper to the type may be accepted for test provided that it can be shown that the absence of the components has no detrimental effect on the results of the test in so far as the requirements of this Regulation are concerned.

3.4.2. It shall be the responsibility of the applicant for approval to show that the application of paragraph 3.4.1. above is compatible with compliance with the requirements of this Regulation.

4. Approval

4.1. If the vehicle type submitted for approval pursuant to this Regulation meets the requirements of this Regulation, approval of that vehicle type shall be granted.

4.1.1. The Technical Service appointed in accordance with paragraph 12. below shall check whether the required conditions have been satisfied.

4.1.2. In case of doubt, account shall be taken, when verifying the conformity of the vehicle to the requirements of this Regulation, of any data or test results provided by the manufacturer which can be taken into consideration in validating the approval test carried out by the Technical Service.

4.2. An approval number shall be assigned to each type approved in accordance with Schedule 4 of the Agreement (E/ECE/TRANS/505/Rev.3).

4.3. Notice of approval or of refusal of approval of a vehicle type pursuant to this Regulation shall be communicated by the Parties to the Agreement which apply this Regulation by means of a form conforming to the model in Annex 1 to this Regulation.

4.4. There shall be affixed, conspicuously and in a readily accessible place specified on the approval form, to every vehicle conforming to a vehicle type approved under this Regulation, an international approval mark consisting of:

4.4.1. A circle surrounding the letter "E" followed by the distinguishing number of the country which has granted approval; [2]

[2]The distinguishing numbers of the Contracting Parties to the 1958 Agreement are reproduced as defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), documentECE/TRANS/WP.29/78/Rev.8, para. 2 - https://unece.org/transport/standards/transport/vehicleregulations-wp29/resolutions

4.4.2. The number of this Regulation, followed by the letter "R", a dash and the approval number, to the right of the circle prescribed in paragraph 4.4.1. above.

4.5. If the vehicle conforms to a vehicle type approved, under one or more other Regulations annexed to the Agreement, in the country which has granted approval under this Regulation, the symbol prescribed in paragraph 4.4.1. above need not be repeated; in such a case the Regulation and approval numbers and the additional symbols of all the Regulations under which approval has been granted in the country which has granted approval under this Regulation shall be placed in vertical columns to the right of the symbol prescribed in paragraph 4.4.1.

4.6. The approval mark shall be clearly legible and be indelible.

4.7. The approval mark shall be placed close to or on the vehicle data plate affixed by the manufacturer.

4.8. Annex 2 to this Regulation gives examples of the arrangements of approval marks.

5. Specifications

5.1. General specifications

5.1.1. The "H" point for each seat shall be determined in accordance with the procedure described in Annex 6.

5.1.2. When the protective system for the front seating positions includes belts, the belt components shall meet the requirements of Regulation No. 16.

5.1.3. Seating positions where a dummy is installed and the protective system includes belts, shall be provided with anchorage points conforming to Regulation No. 14.

5.2. Specifications for the restraint system test (Full Width Rigid Barrier test)

The vehicle shall be tested and approved in accordance with the method described in Annex 3.

The vehicle which, in agreement with the Technical Service, is considered as having the worst case effect on the result of the injury criteria specified in paragraph 5.2.1. shall be selected for this test.

The test of the vehicle carried out in accordance with the method described in Annex 3. shall be considered satisfactory if all the conditions set out in paragraphs 5.2.1. to 5.2.7. below are all satisfied at the same time.

Additionally, vehicles equipped with electric power train shall meet the requirements of paragraph 5.2.8. This can be met by a separate impact test at the request of the manufacturer and after validation by the Technical Service, provided that the electrical components do not influence the occupant protection performance of the vehicle type as defined in paragraphs 5.2.1. to 5.2.5. of this Regulation. In case of this condition the requirements of paragraph 5.2.8. shall be checked in accordance with the methods set out in Annex 3 to this Regulation, except paragraphs 2., 5. and 6. of Annex 3.

A dummy corresponding to the specifications for Hybrid III fiftieth percentile (see footnote 1 of Annex 3) fitted with a 45° ankle angle and meeting the specifications for its adjustment shall be installed in driver's seat.

A dummy corresponding to the specifications for Hybrid III fifth percentile (see footnote 1 of Annex 3) fitted with a 45° ankle angle and meeting the specifications for its adjustment shall be installed in the outboard passenger's seat.

For vehicles of category X:

(a) where the vehicle is equipped to be driven for right hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the left front outboard position and the Hybrid III fifth percentile female dummy on the right front outboard position.

(b) where the vehicle is equipped to be driven for left hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the right front outboard position and the Hybrid III fifth percentile female dummy on the left front outboard position.

5.2.1. The performance criteria described in Annex 4 and recorded in accordance with Annex 8, on the dummies in the front outboard seats shall meet the following conditions:

5.2.1.1. Hybrid III fiftieth percentile adult male performance requirements:

5.2.1.1.1. The head performance criterion (HPC) shall not exceed 1000 and the resultant head acceleration shall not exceed 80 g for more than 3 ms. The latter shall be calculated cumulatively, excluding rebound movement of the head;

5.2.1.1.2. The Injury Criteria for neck shall not exceed the following values:

(a) The axial tensile neck force shall not exceed 3.3 kN;

(b) The fore/aft shear forces at the head/neck interface shall not exceed 3.1 kN;

(c) The neck bending moment about the y axis shall not exceed 57 Nm in extension;

5.2.1.1.3. The thorax compression criterion (ThCC) shall not exceed 42 mm.

5.2.1.1.4. The viscous criterion (V * C) for the thorax shall not exceed 1.0 m/s;

5.2.1.1.5. The femur force criterion (FFC) shall not exceed 9.07 kN.

5.2.1.2. Hybrid III fifth percentile adult female performance requirements:

5.2.1.2.1. The head performance criterion (HPC) shall not exceed 1000 and the resultant head acceleration shall not exceed 80 g for more than 3 ms. The latter shall be calculated cumulatively, excluding rebound movement of the head;

5.2.1.2.2. The Injury Criteria for neck shall not exceed the following values:

(a) The axial tensile neck force shall not exceed 2.9 kN;

(b) The fore/aft shear forces at the head/neck interface shall not exceed 2.7 kN;

(c) The neck bending moment about the y axis shall not exceed 57 Nm in extension;

5.2.1.2.3. The thorax compression criterion (ThCC) shall not exceed:

(a) 34 mm in the case of vehicles of category M_1;

(b) 34 mm in the case of vehicles of category N_1 with a maximum permissible mass not exceeding 2,800 kg, a maximum vehicle width not exceeding 1,480 mm, a length not exceeding 3,400 mm; and

(c) 42 mm in the case of vehicles of category N_1 with a maximum permissible mass exceeding 2,800 kg, or with a maximum vehicle width exceeding 1,480 mm or with a length exceeding 3,400 mm.

5.2.1.2.4. The viscous criterion (V * C) for the thorax shall not exceed 1.0 m/s;

5.2.1.2.5. The femur force criterion (FFC) shall not exceed 7 kN.

5.2.2. Steering Wheel Displacement

5.2.2.1. After the test the residual steering wheel displacement, when measured at the centre of the steering wheel hub, shall not exceed 80 mm in the upwards vertical direction and 100 mm in the rearward horizontal direction.

5.2.2.2. Vehicles meeting the steering wheel displacement requirements of either Regulations Nos. 12 or 94 are deemed to comply with paragraph 5.2.2.1. above.

5.2.3. During the test no door shall open.

5.2.3.1 In the case of automatically activated door locking systems which are installed optionally and/or which can be de-activated by the driver or by the ADS, this requirement shall be verified by using one of the following 2 test procedures, at the choice of the manufacturer:

5.2.3.1.1. If testing in accordance with Annex 3, paragraph 1.4.3.5.2.1., the manufacturer shall in addition demonstrate to the satisfaction of the Technical Service (e.g. manufacturer's in-house data) that, in the absence of the system or when the system is de-activated, no door will open in case of the impact.

5.2.3.1.2. The test is conducted in accordance with Annex 3, paragraph 1.4.3.5.2.2.

5.2.4. After the impact, the side doors shall be unlocked.

5.2.4.1. In the case of vehicles equipped with an automatically activated door locking system, the doors shall be locked before the moment of impact and be unlocked after the impact.

5.2.4.2. In the case of vehicles equipped with automatically activated door locking systems which are installed optionally and/or which can be de-activated by the driver or by the ADS, this requirement shall be verified by using one of the following 2 test procedures, at the choice of the manufacturer:

5.2.4.2.1. If testing in accordance with Annex 3, paragraph 1.4.3.5.2.1., the manufacturer shall in addition demonstrate to the satisfaction of the Technical Service (e.g. manufacturer's in-house data) that, in the absence of the system or when the system is de-activated, no locking of the side doors shall occur during the impact.

5.2.4.2.2. The test is conducted in accordance with Annex 3, paragraph 1.4.3.5.2.2.

5.2.5. After the impact, it shall be possible, without the use of tools except for those necessary to support the weight of the dummy:

5.2.5.1. To open at least one door per row of seats. Where there is no such door, it shall be possible to allow the evacuation of all the occupants by activating the displacement system of seats, if necessary. This is not applicable to convertibles where the top can be easily opened to allow the evacuation of the occupants.

This shall be assessed for all configurations or worst-case configuration for the number of doors on each side of the vehicle and for both left-hand drive and right-hand drive vehicles, when applicable.

5.2.5.2. To release the dummies from their restraint system which, locked, shall be capable of being released by a maximum force of 60 N on the centre of the release control;

5.2.5.3. To remove the dummies from the vehicle without adjustment of the seats.

5.2.6. In the case of a vehicle propelled by liquid fuel, no more than slight leakage of liquid from the fuel feed installation shall occur on collision.

5.2.6.1. If there is continuous leakage of liquid from the fuel-feed installation after the collision, the rate of leakage shall not exceed 30 g/min; if the liquid from the fuel-feed system mixes with liquids from the other systems and the various liquids cannot easily be separated and identified, all the liquids collected shall be taken into account in evaluating the continuous leakage.

5.2.7. In the case of a compressed hydrogen-fuelled vehicle, compliance with paragraphs 5.2.7.1. to 5.2.7.3. shall be shown.

5.2.7.1. The hydrogen leakage rate (V_{H2}) determined in accordance with either, paragraph 4. of Annex 10 for hydrogen, or paragraph 5. of Annex 10 for helium, shall not exceed an average of 118 NL per minute for the time interval, Δt minutes, after the crash.

5.2.7.2. The gas (hydrogen or helium as applicable) concentration by volume in air values determined for the passenger and luggage compartments in accordance with paragraph 6. of Annex 10, shall not exceed 4.0 per cent for hydrogen or 3.0 per cent for helium, at any time throughout the 60 minute post-crash measurement period. This requirement is satisfied if it is confirmed that the shut-off valve of each compressed hydrogen storage system has closed within five seconds of first vehicle contact with the barrier and there is no leakage from the compressed hydrogen storage system(s).

5.2.7.3. The container(s) (for hydrogen storage) shall remain attached to the vehicle at a minimum of one attachment point.

5.2.8. Following the test conducted in accordance with the procedure defined in Annex 3 to this Regulation, the electric power train operating on high voltage and the high voltage systems which are galvanically connected to the high voltage bus of the electric power train shall meet the following requirements:

5.2.8.1. Protection against electrical shock

After the impact, the high voltage buses shall meet at least one of the four criteria specified in paragraph 5.2.8.1.1. through paragraph 5.2.8.1.4.2. below.

If the vehicle has an automatic disconnect function, or device(s) that conductively divide the electric power train circuit during driving condition, at least one of the following criteria shall apply to the disconnected circuit or to each divided circuit individually after the disconnect function is activated.

However, criteria defined in 5.2.8.1.4. below shall not apply if more than a single potential of a part of the high voltage bus is not protected under the conditions of protection degree IPXXB.

In the case that the crash test is performed under the condition that part(s) of the high voltage system are not energized and with the exception of any coupling system for charging the REESS which is not energized during driving condition, the protection against electrical shock shall be proved by either paragraph 5.2.8.1.3. or paragraph 5.2.8.1.4. below for the relevant part(s).

5.2.8.1.1. Absence of high voltage

The voltages U_b, U_1 and U_2 of the high voltage buses shall be equal or less than 30 VAC or 60 VDC within 60 seconds after the impact when measured in accordance with paragraph 2. of Annex 9.

5.2.8.1.2. Low electrical energy

The Total Energy (TE) on the high voltage buses shall be less than 0.2 joules when measured according to the test procedure as specified in paragraph 3. of Annex 9 with the formula (a). Alternatively the total energy (TE) may be calculated by the measured voltage U_b of the high voltage bus and the capacitance of the X-capacitors (C_x) specified by the manufacturer according to formula (b) of paragraph 3. of Annex 9.

The energy stored in the Y-capacitors (TE_{y1}, TE_{y2}) shall also be less than 0.2 joules. This shall be calculated by measuring the voltages U_1 and U_2 of the high voltage buses and the electrical chassis, and the capacitance of the Y-capacitors specified by the manufacturer according to formula (c) of paragraph 3. of Annex 9.

5.2.8.1.3. Physical protection

For protection against direct contact with high voltage live parts, the protection degree IPXXB shall be provided.

The assessment shall be conducted in accordance with paragraph 4 of Annex 9.

In addition, for protection against electrical shock which could arise from indirect contact, the resistance between all exposed conductive parts of electrical protection barriers/enclosures and the electrical chassis shall be lower than 0.1 Ω and the resistance between any two simultaneously reachable exposed conductive parts of electrical protection barriers/enclosures that are less than 2.5 m from each other shall be less than 0.2 Ω when there is current flow of at least 0.2 A. This resistance may be calculated using the separately measured resistances of the relevant parts of electric path.

These requirements are satisfied if the galvanic connection has been made by welding. In case of doubt or the connection is established by mean other than welding, measurements shall be made by using one of the test procedures described in paragraph 4.1. of Annex 9.

5.2.8.1.4. Isolation resistance

The criteria specified in paragraphs 5.2.8.1.4.1. and 5.2.8.1.4.2. below shall be met.

The measurement shall be conducted in accordance with paragraph 5. of Annex 9.

5.2.8.1.4.1. Electric power train consisting of separate DC- or AC-buses

If the AC high voltage buses and the DC high voltage buses are galvanically isolated from each other, isolation resistance between the high voltage bus and the electrical chassis (Ri, as defined in paragraph 5. of Annex 9) shall have a minimum value of 100 Ω/V of the working voltage for DC buses, and a minimum value of 500 Ω/V of the working voltage for AC buses.

5.2.8.1.4.2. Electric power train consisting of combined DC- and AC-buses

If the AC high voltage buses and the DC high voltage buses are conductively connected, they shall meet one of the following requirements:

(a) Isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 500 Ω/V of the working voltage;

(b) Isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 100 Ω/V of the working voltage and the AC bus meets the physical protection as described in paragraph 5.2.8.1.3;

(c) Isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 100 Ω/V of the working voltage and the AC bus meets the absence of high voltage as described in paragraph 5.2.8.1.1.

5.2.8.2. Electrolyte leakage

5.2.8.2.1. In case of aqueous electrolyte REESS.

For a period from the impact until 60 minutes after the impact, there shall be no electrolyte leakage from the REESS into the passenger compartment and no more than 7 per cent by volume of the REESS electrolyte with a maximum of 5.0 l leaked from the REESS to the outside of the passenger compartment. The leaked amount of electrolyte can be measured by usual techniques of determination of liquid volumes after its collection. For containers containing Stoddard, coloured coolant and electrolyte, the fluids shall be allowed to separate by specific gravity then measured.

5.2.8.2.2. In case of non-aqueous electrolyte REESS.

For a period from the impact until 60 minutes after the impact, there shall be no liquid electrolyte leakage from the REESS into the passenger compartment, luggage compartment and no liquid electrolyte leakage to outside the vehicle. This requirement shall be verified by visual inspection without disassembling any part of the vehicle.

5.2.8.3. REESS retention

REESS shall remain attached to the vehicle by at least one component anchorage, bracket, or any structure that transfers loads from REESS to the vehicle structure, and REESS located outside the passenger compartment shall not enter the passenger compartment.

5.2.8.4. REESS fire hazards

For a period from the impact until 60 minutes after the impact, there shall be no evidence of fire or explosion from the REESS.

6. Instructions for users of vehicles equipped with airbags

6.1. For a vehicle fitted with airbag assemblies intended to protect occupants, compliance with paragraphs 8.1.8. to 8.1.9. of UN Regulation No. 16 as amended by the 08 series of amendments, or with paragraphs 8.1.9. to 8.1.10. of UN Regulation No. 16 as amended by the 09 series of amendments or with paragraphs 5.1.9. to 5.1.10. of UN Regulation No. 173 original series, shall be demonstrated as from 1 September 2020 for new vehicle types. Before this date the relevant requirements of the preceding series of amendments apply.

7. Modification and extension of approval of the vehicle type

7.1. Every modification of the vehicle type with regard to this Regulation shall be notified to the Type Approval Authority which approved that vehicle type. The Type Approval Authority may then either:

(a) Decide, in consultation with the manufacturer, that a new type approval is to be granted; or

(b) Apply the procedure contained in paragraph 7.1.1. (Revision) and, if applicable, the procedure contained in paragraph 7.1.2. (Extension).

7.1.1. Revision

When the details recorded in the information documents have changed and the Type Approval Authority considers that the modifications made are unlikely to have appreciable adverse effect, and that in any case the vehicle still meets the requirements, the modification shall be designated a "revision".

In such a case, the Type Approval Authority shall issue the revised pages of the information documents as necessary, marking each revised page to show clearly the nature of the modification and the date of re-issue. A consolidated, updated version of the information documents accompanied by a detailed description of the modification, shall be deemed to meet this requirement.

7.1.2. Extension

The modification shall be designated an "extension" if, in addition to the change of the particulars recorded in the information folder:

(a) Further inspections or tests are required; or

(b) Any information on the communication document (with the exception of its attachments) has changed; or

(c) Approval to a later series of amendments is requested after its entry into force.

7.1.2.1. Require the Technical Service responsible for conducting the tests to carry out a further test, among those described below, according to the nature of the modifications. As this regulation primarily intends to assess the vehicle restraint system, further simplified testing may be conducted using alternative test procedures such as those described in Annex 7.

7.1.2.2. Any modification of the vehicle affecting the general form of the structure of the vehicle and/or any increase in mass greater than 8 per cent which in the judgement of the Type approval Authority would have a marked influence on the results of the tests shall require a repetition of the test as described in Annex 3;

7.1.2.3. If the modifications concern only the interior fittings, if the mass does not increase by more than 8 per cent and if the number of front seats initially provided in the vehicle remains the same, the following shall be

carried out:

7.1.2.3.1. A simplified test as provided for in Annex 7; and/or

7.1.2.3.2. A partial test as defined by the Technical Service in relation to the modifications made.

7.2. Notice of confirmation, extension, or refusal of approval shall be communicated by the procedure specified in paragraph 4.3. above, to the Contracting Parties to the Agreement applying this Regulation. In addition, the index to the information documents and to the test reports, attached to the communication document of Annex 1, shall be amended accordingly to show the date of the most recent revision or extension.

8. Conformity of production

The conformity of production procedures shall comply with those set out in the Agreement, Schedule 1 (E/ECE/TRANS/505/Rev.3), with the following requirements:

8.1. Every vehicle approved under this Regulation shall be manufactured so as to conform to the vehicle type approved and satisfy the requirements set forth in paragraphs 5. and 6.;

8.2. The Type Approval Authority which has granted type approval may at any time verify the conformity control methods applied in each production facility. The normal frequency of these verifications shall be once every two years.

9. Penalties for non-conformity of production

9.1. The approval granted in respect of a vehicle type pursuant to this Regulation may be withdrawn if the requirement laid down in paragraph 7.1. above is not complied with.

9.2. If a Contracting Party to the Agreement applying this Regulation withdraws an approval it has previously granted, it shall forthwith so notify the other Contracting Parties applying this Regulation, by means of a copy of the approval form bearing at the end, in large letters, the signed and dated annotation "APPROVAL WITHDRAWN".

10. Production definitively discontinued

If the holder of the approval completely ceases to manufacture the type of vehicle approved in accordance with the Regulation, he shall so inform the Type Approval Authority which granted the approval. Upon receiving the relevant communication that Type Approval Authority shall inform thereof the other Parties to the Agreement applying this Regulation by means of a copy of the approval form bearing at the end, in large letters, the signed and dated annotation "PRODUCTION DISCONTINUED".

11. Names and addresses of Technical Services responsible for conducting approval tests, and of Type Approval Authorities

The Contracting Parties to the Agreement applying this Regulation shall communicate to the United Nations secretariat the names and addresses of the Technical Services responsible for conducting approval tests, of manufacturers authorized to carry out tests and of the Type Approval Authorities which grant approval and to which forms certifying approval or refusal or withdrawal of approval, issued in other countries, are to be sent.

12. Transitional provisions

12.1. As from the official date of entry into force of the 03 series of amendments, no Contracting Party applying this UN Regulation shall refuse to grant or refuse to accept type-approvals under this UN Regulation as amended by the 03 series of amendments.

12.2. As from 1 September 2027, Contracting Parties applying this UN Regulation shall not be obliged to accept type-approvals of vehicles according to the preceding series of amendments, first issued after 1 September 2027.

12.3. Contracting Parties applying this UN Regulation shall continue to accept type-approvals of vehicles according to the preceding series of amendments, first issued before 1 September 2027 provided the transitional provisions in these respective previous series of amendments foresee this possibility.

12.4. Contracting Parties applying this UN Regulation may grant type-approvals according to any preceding series of amendments to this UN Regulation.

12.5. Contracting Parties applying this UN Regulation shall continue to grant extensions of existing approvals to any preceding series of amendments to this UN Regulation.

12.6. Notwithstanding the transitional provisions above, Contracting Parties who start to apply this Regulation after the date of entry into force of the most recent series of amendments are not obliged to accept type-approvals which were granted in accordance with any of the preceding series of amendments to this Regulation.

12.7. As from the official date of entry into force of the 04 series of amendments, no Contracting Party applying this Regulation shall refuse to grant or refuse to accept type-approvals under this Regulation as amended by the 04 series of amendments.

12.8. As from 1 September 2027, Contracting Parties applying this Regulation shall not be obliged to accept type-approvals of vehicles according to the preceding series of amendments, first issued after 1 September 2027.

12.9. Contracting Parties applying this Regulation shall continue to accept type-approvals of vehicles according to the preceding series of amendments, first issued before 1 September 2027, provided the transitional provisions in these respective previous series of amendments foresee this possibility.

12.10. Contracting Parties applying this Regulation may grant type approvals according to any preceding series of amendments to this Regulation.

12.11. Contracting Parties applying this Regulation shall continue to grant extensions of existing approvals to any preceding series of amendments to this Regulation.

12.12. Notwithstanding the transitional provisions above, Contracting Parties who start to apply this Regulation after the date of entry into force of the most recent series of amendments are not obliged to accept type-approvals which were granted in accordance with any of the preceding series of amendments to this Regulation.

Annex 1 Communication

(Maximum format: A4 (210 x 297 mm))



Issued by: Name of administration:

…….....

…….....

…….....

…….....

Concerning:[2]

Approval granted,

Approval extended,

Approval refused,

Approval withdrawn,

Production definitively discontinued,

of a vehicle type with regard to the protection of the occupants in the event of a frontal collision, pursuant to Regulation No. [137]

Approval No.: ..... Extension No.:.....

---

[1]Distinguishing number of the country which has granted/extended/refused/withdrawn approval (see approval provisions in the Regulation).

---

[2]Strike out what does not apply

1. Trade name or mark of the power-driven vehicle

2. Vehicle type

3. Manufacturer's name and address

4. If applicable, name and address of manufacturer's representative

5. Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials

5.1. Description of the protective system installed in the vehicle

Description of interior arrangements or fittings that might affect the tests .....

5.3. Location of the electrical power source

6. Site of engine: forward/rear/central[2]

---

[2]Strike out what does not apply

7. Drive: front-wheel/rear-wheel[2]

---

[2]Strike out what does not apply

8. Mass of vehicle submitted for testing:

Front axle: .....

Rear axle: .....

Total:.....

9. Vehicle submitted for approval on

10. Technical Service responsible for conducting approval tests

11. Date of report issued by that Service

12. Number of report issued by that Service

13. Approval granted/refused/extended/withdrawn[2]

---

[2]Strike out what does not apply

14. Position of approval mark on vehicle

15. Place

16. Date

17. Signature

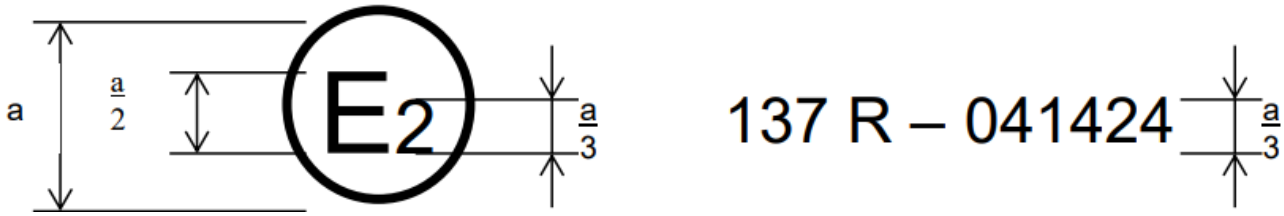
18. The following documents, bearing the approval number shown above, are annexed to this communication:

(Photographs and/or diagrams and drawings permitting the basic identification of the type(s) of vehicle and its possible variants which are covered by the approval)

Annex 2 Arrangements of Approval Marks

Model A

(See paragraph 4.4. of this Regulation.)

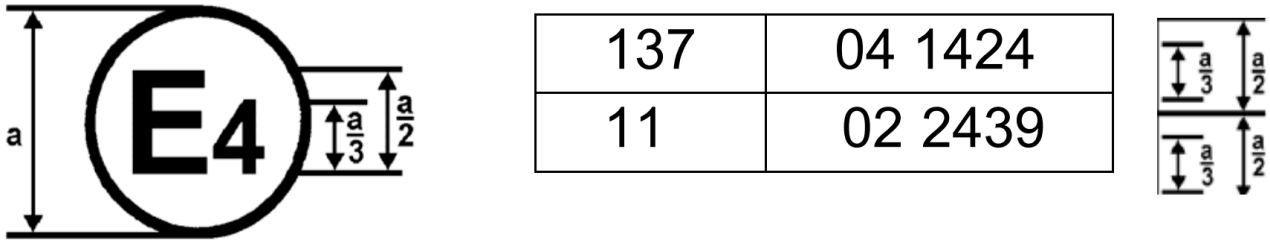


a = 8 mm min.

The above approval mark affixed to a vehicle shows that the vehicle type concerned has, with regard to the protection of the occupants in the event of a frontal collision, been approved in France (E 2) pursuant to Regulation No. 137 under approval number 041424. The approval number indicates that the approval was granted in accordance with the requirements of Regulation No. 137 04 series of amendments.

Model B

(See paragraph 4.5. of this Regulation.)



a = 8 mm min.

The above approval mark affixed to a vehicle shows that the vehicle type concerned has been approved in the Netherlands (E 4) pursuant to Regulations Nos. 137 and 11.[1] The first two digits of the approval numbers indicate that, at the dates when the respective approvals were granted, Regulation No. 137 incorporated the 04 series of amendments and Regulation No. 11 incorporated the 02 series of amendments.

[1] The latter number is given only as an example.

Annex 3 Test procedure

The purpose of this test is to verify whether the vehicle satisfies the requirements set forth in paragraph 5.2.

1. Installation and preparation of the vehicle

1.1. Testing ground

The test area shall be large enough to accommodate the run-up track, barrier and technical installations necessary for the test. The last part of the track, for at least 5 m before the barrier, shall be horizontal, flat and smooth.

1.2. Barrier

The barrier shall consist of a block of reinforced concrete not less than 3 m wide in front and not less than 1.5 m high. The barrier shall be of such thickness that it weighs at least 70 metric tons. The front face shall be flat,

vertical and perpendicular to the axis of the run-up track. It shall be covered with plywood boards 20 ± 2 mm thick, in good condition. A structure on a steel plate at least 25 mm thick may be placed between the plywood board and the barrier. A barrier with different characteristics may likewise be used, provided that the area of the impact surface is greater than the frontal crash area of the vehicle being tested and provided that it gives equivalent results.

1.3. Orientation of the barrier

1.3.1. Alignment of the vehicle to the barrier.

It shall reach the obstacle on a course perpendicular to the collision wall; the maximum lateral misalignment tolerated between the vertical median line of the front of the vehicle and the vertical median line of the collision wall is ± 30 cm.

1.4. State of vehicle

1.4.1. General specification

The test vehicle shall be representative of the series production, shall include all the equipment normally fitted and shall be in normal running order. Some components may be replaced by equivalent masses where this substitution clearly has no noticeable effect on the results measured under paragraph 6.

It shall be allowed by agreement between manufacturer and Technical Service to modify the fuel system so that an appropriate amount of fuel can be used to run the engine or the electrical energy conversion system.

1.4.2. Mass of vehicle

1.4.2.1. For the test, the mass of the vehicle submitted shall be the unladen kerb mass;

1.4.2.2. The fuel tank shall be filled with water to mass equal to 90 per cent of the mass of a full load of fuel as specified by the manufacturer with a tolerance of ±1 per cent;

This requirement does not apply to hydrogen fuel tanks.

1.4.2.3. All the other systems (brake, cooling ...) may be empty in this case, the mass of the liquids shall be compensated;

1.4.2.4. If the mass of the measuring apparatus on board the vehicle exceeds the 25 kg allowed; it may be compensated by reductions which have no noticeable effect on the results measured under paragraph 6 below.

1.4.2.5. The mass of the measuring apparatus shall not change each axle reference load by more than 5 per cent, each variation not exceeding 20 kg.

1.4.2.6. The mass of the vehicle resulting from the provisions of paragraph 1.4.2.1. above shall be indicated in the report.

1.4.3. Passenger compartment adjustments

1.4.3.1. Position of steering wheel

The steering wheel, if adjustable, shall be placed in the normal position indicated by the manufacturer or, in the absence of any particular recommendation by the manufacturer, midway between the limits of its range(s) of adjustment. At the end of the propelled travel, the steering wheel shall be left free, with its spokes in the position which according to the manufacturer corresponds to straight-ahead travel of the vehicle.

1.4.3.2. Glazing

The movable glazing of the vehicle shall be in the closed position. For test measurement purposes and in agreement with the manufacturer, it may be lowered, provided that the position of the operating handle corresponds to the closed position.

1.4.3.3. Gear-change lever position

The gear-change lever shall be in the neutral position. If the vehicle is propelled by its own engine, then the gear-change lever position shall be defined by the manufacturer.

1.4.3.4. Pedals

The pedals shall be in their normal position of rest. If adjustable, they shall be set in their mid-position unless another position is specified by the manufacturer.

1.4.3.5. Doors

The doors shall be closed but not locked.

1.4.3.5.1. In the case of vehicles equipped with an automatically activated door locking system, the system shall be activated at the start of propulsion of the vehicle in order to lock the doors automatically before the moment of impact. At the choice of the manufacturer, the doors shall be locked manually before the start of propulsion of the vehicle.

1.4.3.5.2. In the case of vehicles equipped with an automatically activated door locking system that is installed optionally and/or which can be de-activated by the driver or by the ADS, one of the following two procedures shall be used at the choice of the manufacturer:

1.4.3.5.2.1. The system shall be activated at the start of propulsion of the vehicle in order to lock the doors automatically before the moment of impact. At the choice of the manufacturer, the doors shall be locked manually before the start of propulsion of the vehicle.

1.4.3.5.2.2. The side doors on one side shall be unlocked and the system overridden for these doors; for the side doors on the other vehicle side, the system may be activated in order to lock these doors automatically before the moment of impact. At the choice of the manufacturer, these doors shall be locked manually before the start of propulsion of the vehicle.

1.4.3.6. Opening roof

If an opening or removable roof is fitted, it shall be in place and in the closed position. For test measurement purposes and in agreement with the manufacturer, it may be open.

1.4.3.7. Sun-visor

The sun-visors shall be in the stowed position.

1.4.3.8. Rear-view mirror

The interior rear-view mirror shall be in the normal position of use.

1.4.3.9. Arm-rests

Arm-rests at the front and rear, if movable, shall be in the lowered position, unless this is prevented by the position of the dummies in the vehicles.

1.4.3.10. Head restraints

Head restraints adjustable for height shall be in their appropriate position as defined by the manufacturer. In the absence of any particular recommendation from the manufacturer, then the head restraints shall be in their uppermost position for the 50th percentile male and in the lowermost position for the 5th percentile female dummy.

1.4.3.11. Seats

1.4.3.11.1. Position of front seats

Seats adjustable longitudinally shall be placed so that their "H" point, determined in accordance with the procedure set out in annex 6 is in the middle position of travel or in the nearest locking position thereto, and at the height position defined by the manufacturer (if independently adjustable for height).

In the case of a bench seat, the reference shall be to the "H" point of the driver's place. For vehicles of category X, the "H" point of the most forward outboard passenger’s place shall be taken instead of the H-point of the driver's place.

1.4.3.11.2. Position of front passenger seat

Seats adjustable longitudinally shall be placed so that their "H" point, determined in accordance with the procedure set out in Annex 6, is**:**

(a) In the position given by the manufacturer, which shall be forward of the middle position of travel, or

(b) In the absence of any particular recommendation from the manufacturer, as near as possible to a position which is midway between the forward most position of the seat and the centre position of its travel

Any support system shall be adjusted as defined by the manufacturer. In the absence of any particular recommendation from the manufacturer, then any support system (e.g. seat cushion length and tilt adjustment) shall be in its retracted/ lowermost position.

1.4.3.11.3. Position of the front seat-backs

If adjustable, the seat-backs shall be adjusted so that the resulting inclination of the torso of the dummy is as close as possible to that recommended by the manufacturer for normal use or, in the absence of any particular recommendation by the manufacturer, to 25° towards the rear from the vertical. For the 5th percentile female dummy, the seat back may be adjusted to a different angle, if this is needed to respect the requirements of Annex 5, paragraph 3.1.2.

1.4.3.11.4. Rear seats

If adjustable, the rear seats or rear bench seats shall be placed in the rearmost position.

1.4.4. Electric power train adjustment

1.4.4.1. Procedures for SOC adjustment.

1.4.4.1.1. The adjustment of SOC shall be conducted at an ambient temperature of 20 ± 10 °C.

1.4.4.1.2. The SOC shall be adjusted according to one of the following procedures as applicable. Where different charging procedures are possible, the REESS shall be charged using the procedure which yields the highest SOC:

(a) For a vehicle with a REESS designed to be externally charged, the REESS shall be charged to the highest SOC in accordance with the procedure specified by the manufacturer for normal operation until the charging process is normally terminated.

(b) For a vehicle with a REESS designed to be charged only by an energy source on the vehicle, the REESS shall be charged to the highest SOC which is achievable with normal operation of the vehicle. The manufacturer shall advise on the vehicle operation mode to achieve this SOC.

1.4.4.1.3. When the vehicle is tested, the SOC shall be no less than 95 per cent of the SOC according to paragraphs 1.4.4.1.1. and 1.4.4.1.2. for REESS designed to be externally charged and shall be no less than 90 per cent of SOC according to paragraphs 1.4.4.1.1. and 1.4.4.1.2. for REESS designed to be charged only by an energy source on the vehicle. The SOC will be confirmed by a method provided by the manufacturer.

1.4.4.2. The electric power train shall be energized with or without the operation of the original electrical energy sources (e.g. engine-generator, REESS or electric energy conversion system), however:

1.4.4.2.1. by the agreement between Technical Service and manufacturer it shall be permissible to perform the test with all or parts of the electric power train not being energized insofar as there is no negative influence on the test result. For parts of the electric power train not energized, the protection against electrical shock shall be proved by either physical protection or isolation resistance and appropriate additional evidence.

1.4.4.2.2. in the case where an automatic disconnect is provided, at the request of the manufacturer it shall be permissible to perform the test with the automatic disconnect being triggered. In this case it shall be demonstrated that the automatic disconnect would have operated during the impact test. This includes the automatic activation signal as well as the galvanic separation considering the conditions as seen during the impact.

2. Dummies

2.1. Front seats

2.1.1. A dummy corresponding to the specifications for Hybrid III fiftieth percentile male dummy[1] meeting the specifications for its adjustment shall be installed in the driver seat in accordance with the conditions set out in Annex 5.

A dummy corresponding to the specifications for Hybrid III fifth percentile female dummy[1] meeting the specifications for its adjustment shall be installed in the front passenger seat in accordance with the conditions set out in Annex 5.

For vehicles of category X:

(a) where the vehicle is equipped to be driven in right hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the left front outboard position and the Hybrid III fifth percentile female dummy on the right front outboard position.

(b) where the vehicle is equipped to be driven in left hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the right front outboard position and the Hybrid III fifth percentile female dummy on the left front outboard position.

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[1]The Working Party on Passive Safety (GRSP) of UNECE intends to prepare an addendum for the Mutual Resolution M.R.1 on frontal impact dummies. Until the addendum is available the technical specifications and detailed drawings of Hybrid III with the principal dimensions of a fiftieth percentile male and of a fifth percentile female dummy and the specifications for their adjustment for this test are deposited with the Secretary-General of the United Nations and may be consulted on request at the secretariat to the Economic Commission for Europe, Palais des Nations, Geneva, Switzerland.

2.1.2. The car will be tested with restraint systems, as provided by the manufacturer.

3. Propulsion and course of vehicle

3.1. The vehicle shall be propelled by its own engine or by any other propelling device;

3.2. At the moment of impact the vehicle shall no longer be subject to the action of any additional steering or propelling device.

3.3. The course of the vehicle shall be such that it satisfies the requirements of paragraphs 1.2. and 1.3.1.

4. Test speed

Vehicle speed at the moment of impact shall be 50 -0/ +1 km/h. However, if the test was performed at a higher impact speed and the vehicle met the requirements, the test shall be considered satisfactory.

5. Measurements to be made on dummies in front seats

5.1. All the measurements necessary for the verification of the performance criteria shall be made with measurement systems corresponding to the specifications of annex 8.

5.2. The different parameters shall be recorded through independent data channels of the following CFC (Channel Frequency Class):

5.2.1. Measurements in the head of the dummy

The acceleration (a) referring to the centre of gravity is calculated from the triaxial components of the acceleration measured with a CFC of 1000.

5.2.2. Measurements in the neck of the dummy

5.2.2.1. The axial tensile force and the fore/aft shear force at the neck/head interface are measured with a CFC of 1000.

5.2.2.2. The bending moment about a lateral axis at the neck/head interface are measured with a CFC of 600.

5.2.3. Measurements in the thorax of the dummy

The chest deflection between the sternum and the spine is measured with a CFC of 180.

5.2.4. Measurements in the femur of the dummy

5.2.4.1. The axial compressive force is measured with a CFC of 600.

6. Measurements to be made on the Vehicle

6.1. To enable the simplified test described in annex 7 to be carried out, the deceleration time history of the structure shall be determined on the basis of the value of the longitudinal accelerometers at the base of one of the "B" pillars of the vehicle with a CFC of 180 by means of data channels corresponding to the requirements set out in annex 8;

6.2. The speed time history which will be used in the test procedure described in annex 7 shall be obtained from the longitudinal accelerometer at the "B" pillar.

7. Equivalent procedures

7.1. Alternative procedures may be permitted at the discretion of the Type Approval Authority provided equivalence can be demonstrated. A report shall be attached to the approval documentation describing the method used and the results obtained or the reason for not carrying out the test.

7.2. Responsibility for demonstrating the equivalence of the alternative method shall rest with the manufacturer or his agent wishing to use such a method.

Annex 4 Performance criteria

1. Head Performance Criterion (HPC_{36})

1.1. The Head Performance Criterion (HPC_{36}) is considered to be satisfied when, during the test, there is no contact between the head and any vehicle component.

1.2. If, during the test, there is contact between the head and any vehicle component, a calculation of HPC is made, on the basis of the acceleration (a), measured according to paragraph 5.2.1. of Annex 3, by the following expression:

$$HPC = (t_2 - t_1)\left[\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} a\,dt\right]^{2.5}$$

in which:

1.2.1. The term "a" is the resultant acceleration measured according to paragraph 5.2.1. of Annex 3 and is measured in units of gravity,
g (1 g = 9,81 m/s^2);

1.2.2. If the beginning of the head contact can be determined satisfactorily, t_1 and t_2 are the two time instants, expressed in seconds, defining an interval between the beginning of the head contact and the end of the recording for which the value of HPC is maximum;

1.2.3. If the beginning of the head contact cannot be determined, t_1 and t_2 are the two time instants, expressed in seconds, defining a time interval between the beginning and the end of the recording for which the value of HPC is maximum;

1.2.4. Values of HPC for which the time interval (t_1 - t_2) is greater than 36 ms are ignored for the purposes of calculating the maximum value.

1.3. The value of the resultant head acceleration during forward impact which is exceeded for 3 ms cumulatively is calculated from the resultant head acceleration measured according to paragraph 5.2.1. of Annex 3.

2. Injury Criteria for neck

2.1. These criteria are determined by, the axial tensile force and the fore/aft shear forces at the head/neck interface, expressed in kN and measured according to paragraph 5.2.2. of Annex 3.

2.2. The neck bending moment criterion is determined by the bending moment, expressed in Nm, about a lateral axis at the head/neck interface and measured according to paragraph 5.2.2. of Annex 3.

3. Thorax Compression Criterion (THCC) and Viscous Criterion (V * C)

3.1. The thorax compression criterion is determined by the absolute value of the thorax deformation, expressed in mm and measured according to paragraph 5.2.3. of Annex 3.

3.2. The viscous criterion (V * C) is calculated as the instantaneous product of the compression and the rate of deflection of the sternum, measured according to paragraph 6. of this annex and also paragraph 5.2.3. of Annex 3.

4. Femur Force Criterion (FFC)

4.1. This criterion is determined by the compression load expressed in kN, transmitted axially on each femur of the dummy and measured according to paragraph 5.2.4. of Annex 3.

5. Procedure for calculating the viscous criteria (V * C) for Hybrid III dummy

5.1. The viscous criterion is calculated as the instantaneous product of the compression and the rate of deflection of the sternum. Both are derived from the measurement of sternum deflection.

5.2. The sternum deflection response is filtered once at CFC 180. The compression at time t is calculated from this filtered signal as:

C(t) = D(t) / constant,

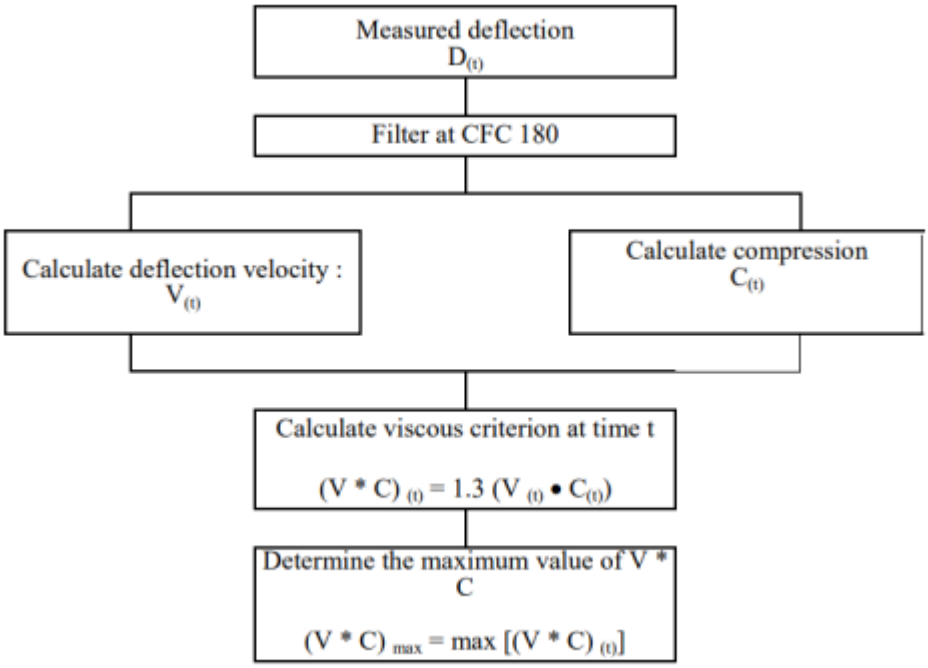
where constant= 0.229 for the HIII 50th

and constant= 0.187 for the HIII 5th

The sternum deflection velocity at time t is calculated from the filtered deflection as:

$$V_{(t)} = \frac{8\left(D_{(t+1)} - D_{(t-1)}\right) - \left(D_{(t+2)} - D_{(t-2)}\right)}{12\partial t}$$

Where $D_{(t)}$ is the deflection at time t in metres and ∂t is the time interval in seconds between the measurements of deflection. The maximum value of ∂t shall be 1.25×10^{-4} seconds. This calculation procedure is shown diagrammatically below:



Annex 5 Arrangement and installation of dummies and adjustment of restraint systems

1. Arrangement of dummies

1.1. Separate seats

The plane of symmetry of the dummy shall coincide with the vertical median plane of the seat.

1.2. Front bench seat

1.2.1. Driver

The plane of symmetry of the dummy shall lie in the vertical plane passing through the steering wheel centre and parallel to the longitudinal median plane of the vehicle. If the seating position is determined by the shape of the bench, such seat shall be regarded as a separate seat.

1.2.2. Outer passenger

The plane of symmetry of the dummy shall be symmetrical with that of the driver dummy relative to the longitudinal median plane of the vehicle. If the seating position is determined by the shape of the bench, such seat shall be regarded as a separate seat.

1.2.3. Outer passengers for vehicles of category X

For vehicles of category X, the plane of symmetry of the dummies shall be vertical and parallel to the vehicle's longitudinal centreline using the Y-coordinate of the R-points as defined by the manufacturer. If the seating position is determined by the shape of the bench, such seat shall be regarded as a separate seat.

1.3. Bench seat for front passengers (not including driver)

The planes of symmetry of the dummy shall coincide with the median planes of the seating positions defined by the manufacturer.

2. Installation of the dummy HIII fiftieth percentile on the driver seat

For vehicles of category X:

(a) where the vehicle is equipped to be driven in right hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the left front outboard position.

(b) where the vehicle is equipped to be driven in left hand traffic, the Hybrid III fiftieth percentile male dummy shall be installed on the right front outboard position.

2.1. Head

The transverse instrumentation platform of the head shall be horizontal within 2.5°. To level the head of the test dummy in vehicles with upright seats with non-adjustable backs, the following sequences shall be followed. First adjust the position of the "H" point within the limits set forth in paragraph 2.4.3.1. below to level the transverse instrumentation platform of the head of the test dummy. If the transverse instrumentation platform of the head is still not level, then adjust the pelvic angle of the test dummy within the limits provided in paragraph 2.4.3.2. below. If the transverse instrumentation platform of the head is still not level, then adjust the neck bracket of the test dummy the minimum amount necessary to ensure that the transverse instrumentation platform of the head is horizontal within 2.5°.

2.2. Arms

2.2.1. The driver's upper arms shall be adjacent to the torso with the centrelines as close to a vertical plane as possible.

2.2.2. For vehicles of category X, the dummy's upper arms shall be in contact with the seat back and the sides of the torso.

2.3. Hands

2.3.1. The palms of the driver test dummy shall be in contact with the outer part of the steering wheel rim at the rim's horizontal centreline. The thumbs shall be over the steering wheel rim and shall be lightly taped to the steering wheel rim so that if the hand of the test dummy is pushed upward by a force of not less than 9 N and not more than 22 N, the tape shall release the hand from the steering wheel rim.

2.3.2. For vehicles of category X, the palms of the test dummy shall be in contact with outside of thigh. The little finger shall be in contact with the seat cushion.

2.4. Torso

2.4.1. In vehicles equipped with bench seats, the upper torso of the driver test dummy shall rest against the seat back. The midsagittal plane of the driver dummy shall be vertical and parallel to the vehicle's longitudinal centreline, and pass through the centre of the steering wheel rim.

For vehicles of category X, the midsagittal plane of the dummy shall be vertical and parallel to the vehicle's longitudinal centreline using the Y-coordinate of the R-point as defined by the manufacturer.

2.4.2. In vehicles equipped with individual seats, the upper torso of the driver test dummy shall rest against the seat back. The midsagittal plane of the driver dummy shall be vertical and shall coincide with the longitudinal centreline of the individual seat.

For vehicles of category X, the midsagittal plane of the dummy shall be vertical and shall coincide with the longitudinal centreline of the individual seat.

2.4.3. Lower torso

2.4.3.1. "H" point

The "H" point of the driver test dummy shall coincide within 13 mm in the vertical dimension and 13 mm in the horizontal dimension, with a point 6 mm below the position of the "H" point determined using the procedure described in Annex 6 except that the length of the lower leg and thigh segments of the "H" point machine shall be adjusted to 414 and 401 mm, instead of 417 and 432 mm respectively.

2.4.3.2. Pelvic angle

As determined using the pelvic angle gauge (GM) drawing 78051-532 incorporated by reference in Part 572 which is inserted into the "H" point gauging hole of the dummy, the angle measured from the horizontal on the

76.2 mm (3 inch) flat surface of the gauge shall be 22.5 degrees plus or minus 2.5 degrees.

2.5. Legs

The upper legs of the driver test dummy shall rest against the seat cushion to the extent permitted by placement of the feet. The initial distance between the outboard knee clevis flange surfaces shall be 270 mm ± 10 mm. To the extent practicable, the left leg of the driver dummy shall be in vertical longitudinal planes. To the extent practicable, the right leg of the driver dummy shall be in a vertical plane. Final adjustment to accommodate placement of feet in accordance with paragraph 2.6. for various passenger compartment configurations is permitted.

For vehicles of category X, the upper legs of the test dummy shall rest against the seat cushion to the extent permitted by placement of the feet. The initial distance between the outboard knee clevis flange surfaces shall be 270 mm ± 10 mm. To the extent practicable, both legs of the test dummy shall be in vertical longitudinal planes. Final adjustment to accommodate placement of feet in accordance with paragraph 2.6. for various passenger compartment configurations is permitted.

2.6. Feet

2.6.1. The right foot of the driver test dummy shall rest on the undepressed accelerator with the rearmost point of the heel on the floor surface in the plane of the pedal. If the foot cannot be placed on the accelerator pedal, it shall be positioned perpendicular to the tibia and placed as far forward as possible in the direction of the centreline of the pedal with the rearmost point of the heel resting on the floor surface. The heel of the left foot shall be placed as far forward as possible and shall rest on the floor pan. The left foot shall be positioned as flat as possible on the toe board. The longitudinal centreline of the left foot shall be placed as parallel as possible to the longitudinal centreline of the vehicle. For vehicles equipped with a footrest, it shall be possible at the request of the manufacturer to place the left foot on the footrest. In this case the position of the left foot is defined by the footrest.

2.6.2. For vehicles of category X, the heels of both feet of the test dummy shall be placed as far forward as possible and shall rest on the floor pan. Both feet shall be positioned as flat as possible on the toe board. The longitudinal centreline of the feet shall be placed as parallel as possible to the longitudinal centreline of the vehicle.

2.7. The measuring instruments installed shall not in any way affect the movement of the dummy during impact.

2.8. The temperature of the dummy and the system of measuring instruments shall be stabilized before the test and maintained so far as possible within a range between 19 °C and 22.2 °C.

2.9. Dummy HIII fiftieth percentile clothing

2.9.1. The instrumented dummy will be clothed in formfitting cotton stretch garments with short sleeves and mid-calf length trousers specified in FMVSS 208, drawings 78051-292 and 293 or their equivalent.

2.9.2. A size 11XW shoe, which meets the configuration size, sole and heel thickness specifications of the United States of America military standard MIL S 13192, revision P and whose weight is 0.57 ± 0.1 kg, shall be placed and fastened on each foot of the test dummy.

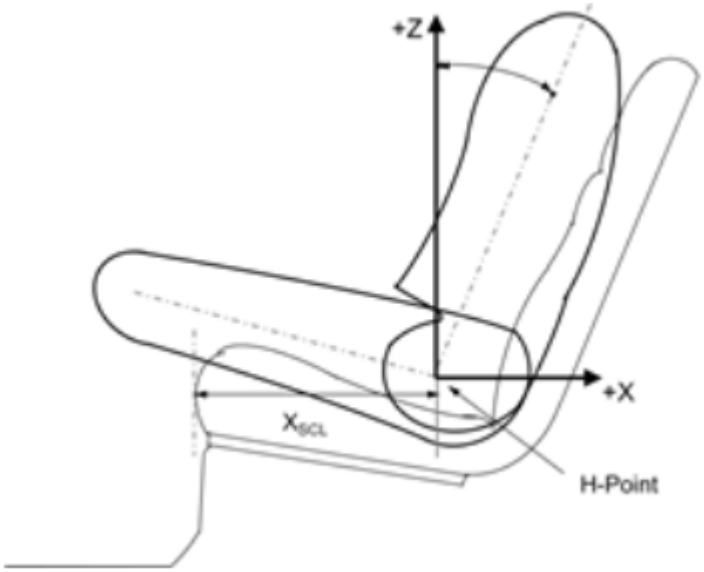
3. Installation of the dummy Hybrid III fifth percentile female dummy on the passenger seat.

The longitudinal and vertical dimension of "H" point are described as (X_{50thM}, Z_{50thM}) and the longitudinal and vertical dimension of "H 5th" point are described as (X_{5thF}, Z_{5thF}). XSCL is defined as the horizontal distance between the "H" point and the most forward point on the seat cushion (see Fig. 1). Use the following formula to calculate the "H 5th" point. Note that X_{5thF} should always be more forward than the X_{50thM}.

$X_{5thF} = X_{50thM}$, + (93mm - 0.323 x XSCL)

$Z_{5thF} = Z_{50thM}$

Figure 1



For vehicles of category X:

(a) where the vehicle is equipped to be driven in right hand traffic, the Hybrid III fifth percentile female dummy shall be on the right front outboard position.

(b) where the vehicle is equipped to be driven in left hand traffic, the Hybrid III fifth percentile female dummy shall be on the left front outboard position.

3.1. Head

The transverse instrumentation platform of the head shall be horizontal within 2.5°.

3.1.1. To level the head of the test dummy in vehicles with upright seats with non-adjustable backs, the following sequences shall be followed. First adjust the position of the "H 5th" point within the limits set forth in paragraph 3.4.3.1. below to level the transverse instrumentation platform of the head of the test dummy. If the transverse instrumentation platform of the head is still not level, then adjust the pelvic angle of the test dummy within the limits provided in paragraph 3.4.3.2. below. If the transverse instrumentation platform of the head is still not level, then adjust the neck bracket of the test dummy to the minimum amount necessary to ensure that the transverse instrumentation platform of the head is horizontal within 2.5°.

3.1.2. To level the head of the test dummy in vehicles with adjustable backs, the following sequences shall be followed. First adjust the position of the "H 5th" point within the limits set forth in paragraph 3.4.3.1. below to level the transverse instrumentation platform of the head of the test dummy. If the transverse instrumentation platform of the head is still not level, then adjust the pelvic angle of the test dummy within the limits provided in paragraph 3.4.3.2. below. If the transverse instrumentation platform of the head is still not level, then adjust the neck bracket of the test dummy to the minimum amount necessary to ensure that the transverse instrumentation platform of the head is horizontal within 2.5°. If the transverse instrumentation platform of the head is still not level, then adjust the seat back to a different angle at a minimum amount necessary to ensure that the transverse instrumentation platform of the head is horizontal within 2.5°.

3.2. Arms

3.2.1. The passenger's upper arms shall be in contact with the seat back and the sides of the torso.

3.3. Hands

3.3.1. The palms of the passenger test dummy shall be in contact with outside of thigh. The little finger shall be in contact with the seat cushion.

3.4. Torso

3.4.1. In vehicles equipped with bench seats, the upper torso of the passenger test dummy shall rest against the seat back. The midsagittal plane of the passenger dummy shall be vertical and parallel to the vehicle's longitudinal centreline and the same distance from the vehicle's longitudinal centreline as the midsagittal plane of the driver dummy.

For vehicles of category X, the midsagittal plane of the Hybrid III fifth percentile female dummy shall be vertical and parallel to the vehicle's longitudinal centreline and the same distance from the vehicle's longitudinal centreline as the midsagittal plane of the Hybrid III fiftieth percentile male dummy.

3.4.2. In vehicles equipped with individual seats, the upper torso of the passenger test dummy shall rest against the seat back. The midsagittal plane of the passenger dummy shall be vertical and shall coincide with the longitudinal centreline of the individual seat.

3.4.3. Lower torso

3.4.3.1. "H 5th " point

The "H 5th" point of passenger test dummy shall coincide within 13 mm in the horizontal dimension, of the "H 5th" point determined using the procedure described in Annex 6 and paragraph 3 above.

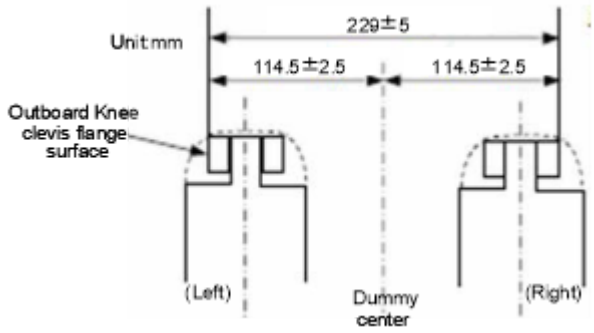
3.4.3.2. Pelvic angle

As determined using the pelvic angle gauge (GM) drawing 78051-532 incorporated by reference in Part 572 which is inserted into the "H" point gauging hole of the dummy, the angle measured from the horizontal on the 76.2 mm (3 inch) flat surface of the gauge shall be 20 degrees plus or minus 2.5 degrees.

3.5. Legs

The upper legs of the passenger test dummy shall rest against the seat cushion to the extent permitted by placement of the feet. The initial distance between the outboard knee clevis flange surfaces shall be 229 mm ± 5 mm as shown in Figure 2. To the extent practicable, both legs of the passenger dummy shall be in vertical longitudinal planes. Final adjustment to accommodate placement of feet in accordance with paragraph 3.6. for various passenger compartment configurations is permitted.

Figure 2 **The initial knee distance of Hybrid III fifth percentile female**



3.6. Feet

3.6.1. The legs shall be positioned as distant as possible from the front end of the seat cushion while the thighs are kept in contact with the seat cushion as shown in Fig.(a). As shown in Fig.(b), each leg shall be lowered until the foot comes in contact with the floor while the foot and tibia are kept in a right angle to one another and the thigh inclination angle kept constant. When each heel is in contact with the floor, the foot shall be rotated so that the toe comes as much in contact as possible with the floor as shown in Fig.(c).

If it is not possible to have each foot in contact with the floor, the foot shall be lowered until the calf comes in contact with the front end of the seat cushion or the back of the foot comes in contact with the vehicle interior. The foot shall be kept as parallel as possible to the floor as shown in Fig.(d).

In case of interference from a vehicle body protrusion, the foot shall be rotated as minimally as possible around the tibia. In case interference still remains, the femur shall be rotated to resolve or minimize the interference. The foot shall be moved inward or outward while the separation distance between the knees is kept constant.

Figure (a)

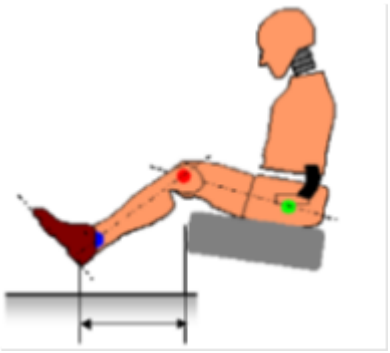


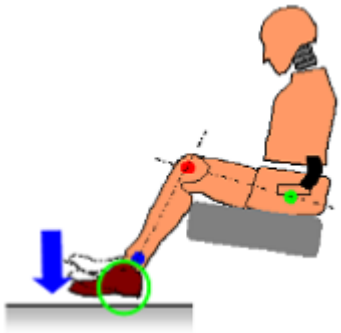
Figure (b)



Figure (c)



Figure (d)



3.7. The measuring instruments installed shall not in any way affect the movement of the dummy during impact.

3.8. The temperature of the dummies and the system of measuring instruments shall be stabilized before the test and maintained so far as possible within a range between 19 °C and 22.2 °C.

3.9. Dummy HIII fifth percentile clothing

3.9.1. The instrumented dummy will be clothed in formfitting cotton stretch garments with short sleeves and mid-calf length trousers specified in FMVSS 208, drawings 78051-292 and 293 or their equivalent.

3.9.2. A size 7.5 W small female size shoe, which meets the configuration size, sole and heel thickness specifications of the United States of America military standard MIL-S-21711E, revision P and whose weight is 0.41 ± 0.09 kg, shall be placed and fastened on each foot of the test dummies.

4. Adjustment of restraint system

The dummy jacket shall be installed at the appropriate position where the bolt hole of the neck lower bracket and the work hole of the dummy jacket should be at the same position. With the test dummy at its designated seating position as specified by the appropriate requirements of paragraphs 2.1. through 2.6. and paragraphs 3.1 to 3.6 above, place the belt around the test dummy and fasten the latch. Remove all slack from the lap belt. Pull the upper torso webbing out of the retractor and allow it to retract. Repeat this operation four times. The shoulder belt should be at the position between the area which shall not be taken off of shoulder and shall not come in contact with the neck. The seat belt path shall be positioned: for Hybrid III fiftieth percentile male dummy, the hole of the

outer side dummy jacket shall not be fully hidden by the seat belt. For Hybrid III fifth percentile female dummy, the seat belt shall lie between the breasts. Apply a 9 to 18 N tension load to the lap belt. If the belt system is equipped with a tension-relieving device, introduce the maximum amount of slack into the upper torso belt that is recommended by the manufacturer for normal use in the owner's manual for the vehicle. If the belt system is not equipped with a tension-relieving device, allow the excess webbing in the shoulder belt to be retracted by the rewind force of the retractor. Where the safety belt and safety belt anchorages are located such that the belt does not lie as required above then the safety belt may be manually adjusted and retained by tape.

Annex 6 Procedure for determining the "H" point and the actual torso angle for seating positions in motor vehicles 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5);

see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 6 Appendix 1 - Description of the three dimensional "H" point machine (3-D H machine) 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5);

see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 6 Appendix 2 - Three - dimensional reference system 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5);

see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 6 Appendix 3 - Reference data concerning seating positions 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5);

see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 7 Test procedure with trolley

1. Test installation and procedure

1.1. Trolley

The trolley shall be so constructed that no permanent deformation appears after the test. It shall be so guided that, during the impact phase, the deviation in the vertical plane does not exceed 5° and 2° in the horizontal plane.

1.2. State of the structure

1.2.1. General

The structure tested shall be representative of the series production of the vehicles concerned. Some components may be replaced or removed where such replacement or removal clearly has no effect on the test results.

1.2.2. Adjustments

Adjustments shall conform to those set out in paragraph 1.4.3. of Annex 3 to this Regulation, taking into account what is stated in paragraph 1.2.1. above.

1.3. Attachment of the structure

1.3.1. The structure shall be firmly attached to the trolley in such a way that no relative displacement occurs during the test.

1.3.2. The method used to fasten the structure to the trolley shall not have the effect of strengthening the seat anchorages or restraint devices, or of producing any abnormal deformation of the structure.

1.3.3. The attachment device recommended is that whereby the structure rests on supports placed approximately in the axis of the wheels or, if possible, whereby the structure is secured to the trolley by the fastenings of the suspension system.

1.3.4. The angle between the longitudinal axis of the vehicle and the direction of motion of the trolley shall be 0° ± 2°.

1.4. Dummies

The dummies and their positioning shall conform to the specifications in Annex 3, paragraph 2.

1.5. Measuring apparatus

1.5.1. Deceleration of the structure

The position of the transducers measuring the deceleration of the structure during the impact shall be parallel to the longitudinal axis of the trolley according to the specifications of Annex 8 (CFC 180).

1.5.2. Measurements to be made on the dummies

All the measurements necessary for checking the listed criteria are set out in Annex 3, paragraph 5.

1.6. Deceleration curve of the structure

The deceleration curve of the structure during the impact phase shall be such that the "variation of speed in relation to time" curve obtained by integration at no point differs by more than ±1 m/s from the "variation of speed in relation to time" reference curve of the vehicle concerned as defined in appendix to this annex. A displacement with regard to the time axis of the reference curve may be used to obtain the structure velocity inside the corridor.

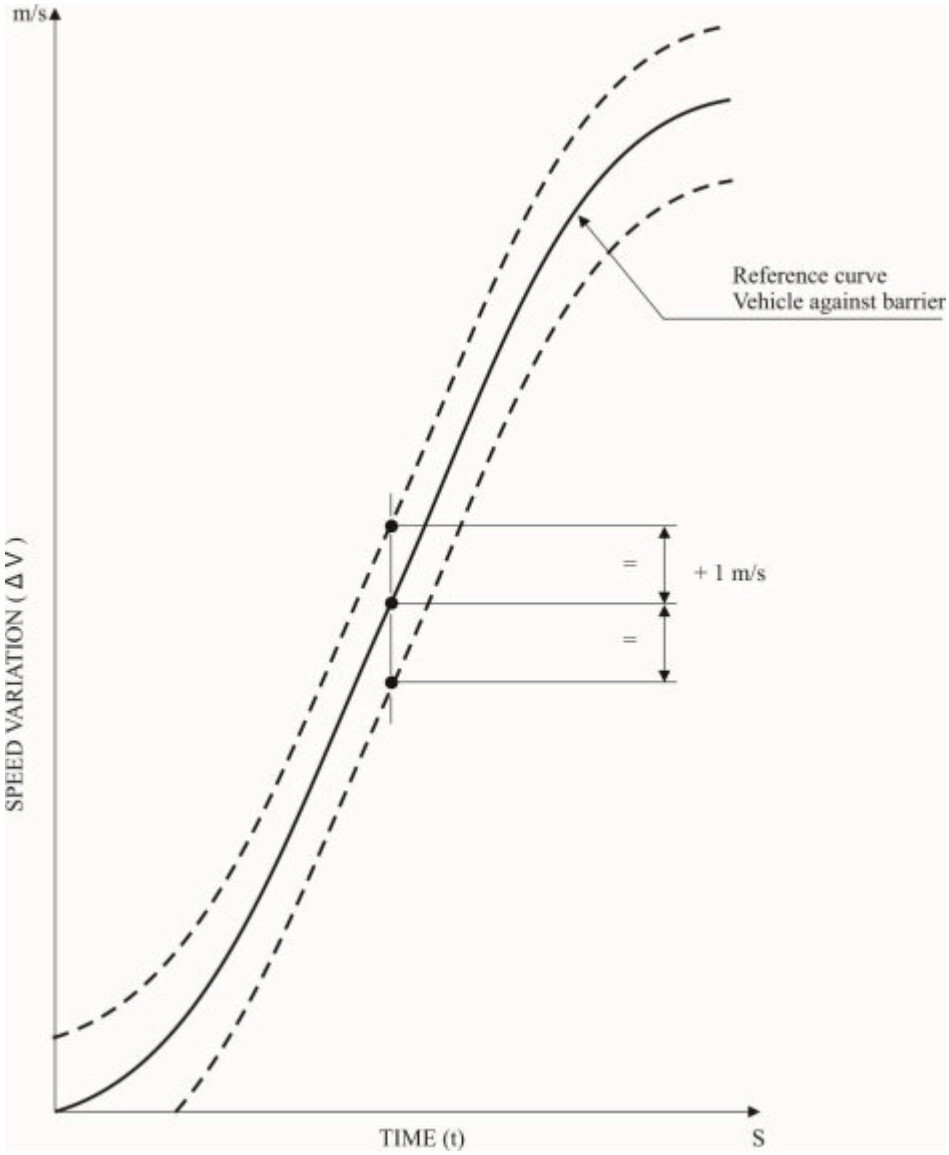
1.7. Reference curve $\Delta V = f(t)$ of the vehicle concerned

This reference curve is obtained by integration of the deceleration curve of the vehicle concerned measured in the frontal collision test against a barrier as provided for in paragraph 6. of Annex 3 to this Regulation.

1.8. Equivalent method

The test may be performed by some other method than that of deceleration of a trolley, provided that such method complies with the requirement concerning the range of variation of speed described in paragraph 1.6. above.

Annex 7 - Appendix Equivalence curve - tolerance band for curve ΔV = f(t)



Annex 8 Technique of measurement in measurement tests: Instrumentation

1. Definitions

1.1. Data channel

A data channel comprises all the instrumentation from a transducer (or multiple transducers whose outputs are combined in some specified way) up to and including any analysis procedures that may alter the frequency content or the amplitude content of data.

1.2. Transducer

The first device in a data channel used to convert a physical quantity to be measured into a second quantity (such as an electrical voltage) which can be processed by the remainder of the channel.

1.3. Channel Amplitude Class: CAC

The designation for a data channel that meets certain amplitude characteristics as specified in this annex. The CAC number is numerically equal to the upper limit of the measurement range.

1.4. Characteristic frequencies F_H, F_L, F_N

These frequencies are defined in Figure 1 of this annex.

1.5. Channels Frequency Class: CFC

The channel frequency class is designated by a number indicating that the channel frequency response lies within the limits specified in Figure 1 of this annex. This number and the value of the frequency F_H in Hz are numerically equal.

1.6. Sensitivity coefficient

The slope of the straight line representing the best fit to the calibration values determined by the method of least square within the channel amplitude class.

1.7. Calibration factor of a data channel

The mean value of the sensitivity coefficients evaluated over frequencies which are evenly spaced on a logarithmic scale between F_L and $\frac{F_H}{2.5}$

1.8. Linearity error

The ratio, in per cent, of the maximum difference between the calibration value and the corresponding value read on the straight line defined in paragraph 1.6. above at the upper limit of the channel amplitude class.

1.9. Cross sensitivity

The ratio of the output signal to the input signal, when an excitation is applied to the transducer perpendicular to the measurement axis. It is expressed as a percentage of the sensitivity along the measurement axis.

1.10. Phase delay time

The phase delay time of a data channel is equal to the phase delay (in radians) of a sinusoidal signal, divided by the angular frequency of that signal (in radians/second).

1.11. Environment

The aggregate, at a given moment, of all external conditions and influences to which the data channel is subjected.

2. Performance requirements

2.1. Linearity error

The absolute value of the linearity error of a data channel at any frequency in the CFC, shall be equal to or less than 2.5 per cent of the value of the CAC, over the whole measurement range.

2.2. Amplitude against frequency

The frequency response of a data channel shall lie within the limiting curves given in Figure 1 of this annex. The zero dB line is determined by the calibration factor.

2.3. Phase delay time

The phase delay time between the input and the output signals of a data channel shall be determined and shall not vary by more than 1/10 F_H seconds between 0.03 F_H and F_H.

2.4. Time

2.4.1. Time base

A time base shall be recorded and shall at least give 1/100 s with an accuracy of 1 per cent.

2.4.2. Relative time delay

The relative time delay between the signal of two or more data channels, regardless of their frequency class, shall not exceed 1 ms excluding delay caused by phase shift.

Two or more data channels of which the signals are combined shall have the same frequency class and shall not have relative time delay greater than 1/10 F_H seconds.

This requirement applies to analogue signals as well as to synchronization pulses and digital signals.

2.5. Transducer cross sensitivity

The transducer cross sensitivity shall be less than 5 per cent in any direction.

2.6. Calibration

2.6.1. General

A data channel shall be calibrated at least once a year against reference equipment traceable to known standards. The methods used to carry out a comparison with reference equipment shall not introduce an error greater than 1 per cent of the CAC. The use of the reference equipment is limited to the frequency range for which they have been calibrated. Subsystems of a data channel may be evaluated individually and the results factored into the accuracy of the total data channel. This can be done for example by an electrical signal of known amplitude simulating the output signal of the transducer which allows a check to be made on the gain factor of the data channel, excluding the transducer.

2.6.2. Accuracy of reference equipment for calibration

The accuracy of the reference equipment shall be certified or endorsed by an official metrology service.

2.6.2.1. Static calibration

2.6.2.1.1. Accelerations

The errors shall be less than ±1.5 per cent of the channel amplitude class.

2.6.2.1.2. Forces

The error shall be less than ±1 per cent of the channel amplitude class.

2.6.2.1.3. Displacements

The error shall be less than ±1 per cent of the channel amplitude class.

2.6.2.2. Dynamic calibration

2.6.2.2.1. Accelerations

The error in the reference accelerations expressed as a percentage of the channel amplitude class shall be less than ±1.5 per cent below 400 Hz, less than ±2 per cent between 400 Hz and 900 Hz, and less than ±2.5 per cent above 900 Hz.

2.6.2.3. Time

The relative error in the reference time shall be less than 10^{-5}.

2.6.3. Sensitivity coefficient and linearity error

The sensitivity coefficient and the linearity error shall be determined by measuring the output signal of the data channel against a known input signal for various values of this signal. The calibration of the data channel shall cover the whole range of the amplitude class.

For bi-directional channels, both the positive and negative values shall be used. If the calibration equipment cannot produce the required input owing to the excessively high values of the quantity to be measured, calibrations shall be carried out within the limits of the calibration standards and these limits shall be recorded in the test report.

A total data channel shall be calibrated at a frequency or at a spectrum of frequencies having a significant value between F_L and $\frac{F_H}{2.5}$.

2.6.4. Calibration of the frequency response

The response curves of phase and amplitude against frequency shall be determined by measuring the output signals of the data channel in terms of phase and amplitude against a known input signal, for various values of this signal varying between F_L and 10 times the CFC or 3,000 Hz, whichever is lower.

2.7. Environmental effects

A regular check shall be made to identify any environmental influence (such as electric or magnetic flux, cable velocity, etc.). This can be done for instance by recording the output of spare channels equipped with dummy transducers. If significant output signals are obtained corrective action shall be taken, for instance by replacement of cables.

2.8. Choice and designation of the data channel

The CAC and CFC define a data channel.

The CAC shall be 1, 2 or 5 to a power of ten.

3. Mounting of transducers

Transducers should be rigidly secured so that their recordings are affected by vibration as little as possible. Any mounting having a lowest resonance frequency equal to at least 5 times the frequency F_H of the data channel considered shall be considered valid. Acceleration transducers in particular should be mounted in such a way that the initial angle of the real measurement axis to the corresponding axis of the reference axis system is not greater than 5° unless an analytical or experimental assessment of the effect of the mounting on the collected data is made. When multi-axial accelerations at a point are to be measured, each acceleration transducer axis should pass within 10 mm of that point, and the centre of seismic mass of each accelerometer should be within 30 mm of that point.

4. Data processing

4.1. Filtering

Filtering corresponding to the frequencies of the data channel class may be carried out during either recording or processing of data. However, before recording, analogical filtering at a higher level than CFC should be effected in order to use at least 50 per cent of the dynamic range of the recorder and to reduce the risk of high frequencies saturating the recorder or causing aliasing errors in the digitalizing process.

4.2. Digitalizing

4.2.1. Sampling frequency

The sampling frequency should be equal to at least 8 F_H.

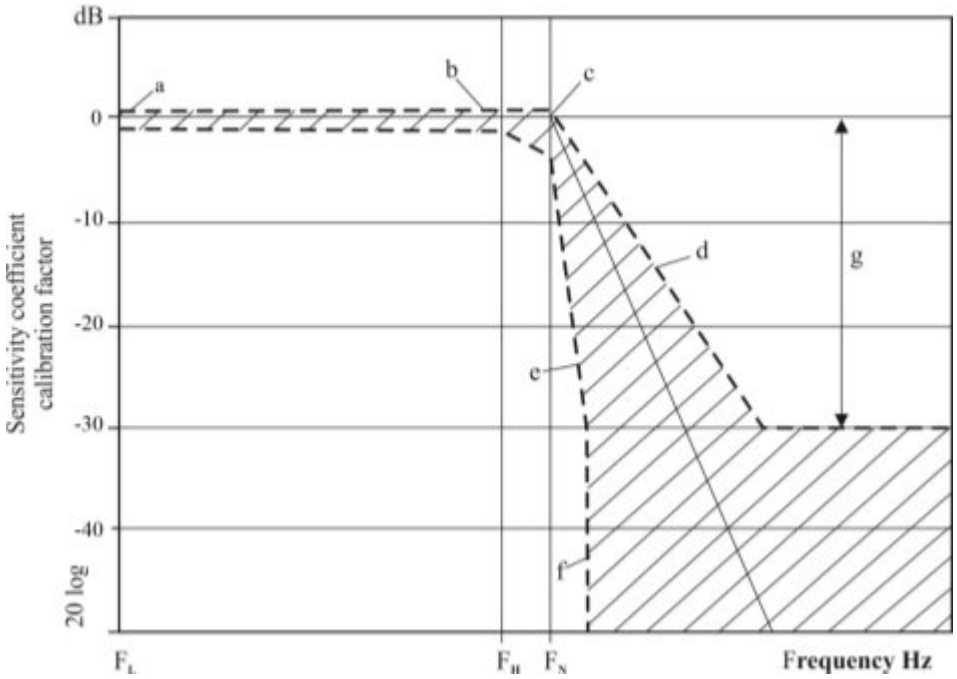
4.2.2. Amplitude resolution

The size of digital words should be at least 7 bits and a parity bit.

5. Presentation of results

The results should be presented on A4 size paper (ISO/R 216). Results presented as diagrams should have axes scaled with a measurement unit corresponding to a suitable multiple of the chosen unit (for example, 1, 2, 5, 10, 20 millimetres). SI units shall be used, except for vehicle velocity, where km/h may be used, and for accelerations due to impact where g, with g = 9.8 m/s^2, may be used.

Figure 1 **Frequency response curve**



CFC	F_L Hz	F_H Hz	F_N Hz	N Logarithmic scale
				a ± 0.5 dB
				b + 0.5; -1 dB
1,000	< 0.1	1,000	1,650	c + 0.5; -4 dB
600	< 0.1	600	1,000	d - 9 dB/octave
180	< 0.1	180	300	e - 24 dB/octave
60	< 0.1	60	100	f ∞
				g - 30

Annex 9 Test Procedures for vehicles equipped with the electric power train

This annex describes test procedures to demonstrate compliance to the electrical safety requirements of paragraph 5.2.8. of this Regulation.

1. Test setup and equipment

If a high voltage disconnect function is used, measurements are to be taken from both sides of the device performing the disconnect function. However, if the high voltage disconnect is integral to the REESS or the energy conversion system and the high-voltage bus of the REESS or the energy conversion system is protected according to protection IPXXB following the impact test, measurements may only be taken between the device performing the disconnect function and the electrical loads. The voltmeter used in this test shall measure DC values and have an internal resistance of at least 10 MΩ.

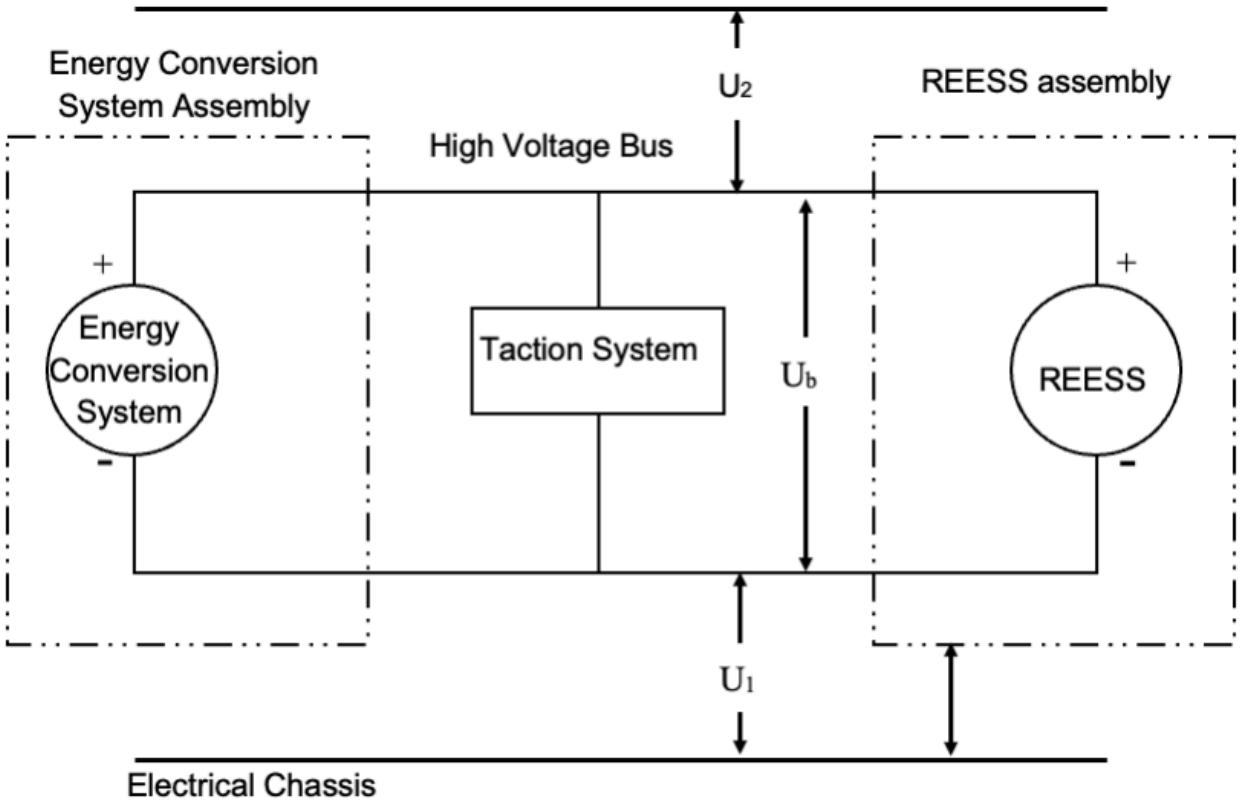
2. The following instructions may be used if voltage is measured.

After the impact test, determine the high voltage bus voltages (U_b, U_1, U_2) (see Figure 1 below).

The voltage measurement shall be made not earlier than 10 seconds, but, not later than 60 seconds after the impact.

This procedure is not applicable if the test is performed under the condition where the electric power train is not energized.

Figure 1 **Measurement of U_b, U_1, U_2**



3. Assessment procedure for low electrical energy

Prior to the impact a switch S_1 and a known discharge resistor R_e is connected in parallel to the relevant capacitance (ref. Figure 2 below).

(a) Not earlier than 10 seconds and not later than 60 seconds after the impact the switch S1 shall be closed while the voltage U_b and the current Ie are measured and recorded. The product of the voltage U_b and the current Ie shall be integrated over the period of time, starting from the moment when the switch S1 is closed (tc) until the voltage U_b falls below the high voltage threshold of 60 V DC (th). The resulting integration equals the Total Energy (TE) in joules.

$$TE = \int_{tc}^{th} U_b \times I_e dt$$

(b) When U_b is measured at a point in time between 10 seconds and
60 seconds after the impact and the capacitance of the X-capacitors (C_x) is specified by the manufacturer, Total Energy (TE) shall be calculated according to the following formula:

$TE = 0.5 \times C_x \times U_b^2$

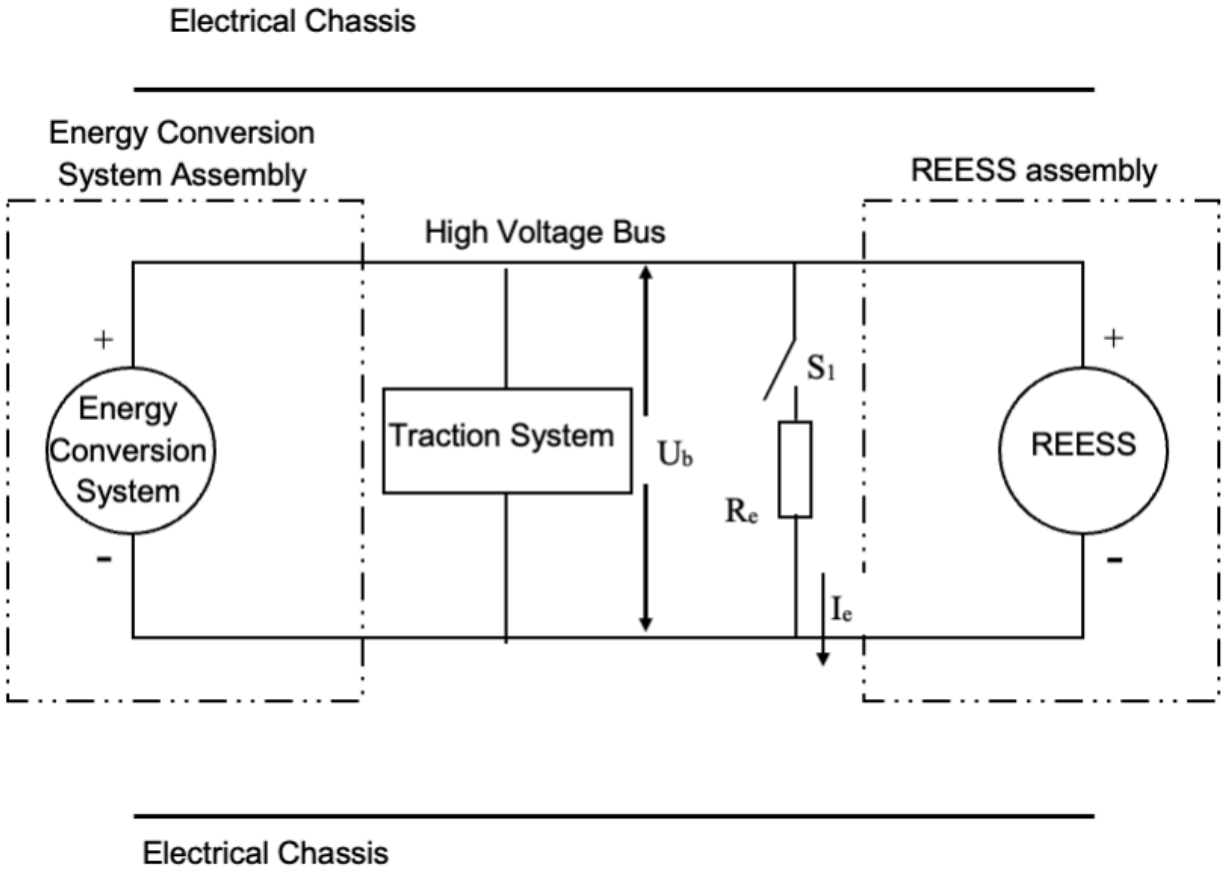
(c) WhenU_1 and U_2 (see Figure 1 above) are measured at a point in time between 10 seconds and 60 seconds after the impact and the capacitances of the Y-capacitors (C_{y1}, C_{y2}) are specified by the manufacturer, Total Energy (TE_{y1}, TE_{y2}) shall be calculated according to the following formulas:

$TE_{y1} = 0.5 \times C_{y1} \times U_1^2$

$TE_{y2} = 0.5 \times C_{y2} \times U_2^2$

This procedure is not applicable if the test is performed under the condition where the electric power train is not energized.

Figure 2 **E.g. measurement of high voltage bus energy stored in X-capacitors**



4. Physical protection

Following the vehicle impact test any parts surrounding the high voltage components shall be, without the use of tools, opened, disassembled or removed. All remaining surrounding parts shall be considered part of the physical protection.

The jointed test finger described in Figure 3 shall be inserted into any gaps or openings of the physical protection with a test force of 10 N ± 10 per cent for electrical safety assessment. If partial or full penetration into the physical protection by the jointed test finger occurs, the jointed test finger shall be placed in every position as specified below.

Starting from the straight position, both joints of the test finger shall be rotated progressively through an angle of up to 90 degrees with respect to the axis of the adjoining section of the finger and shall be placed in every possible position.

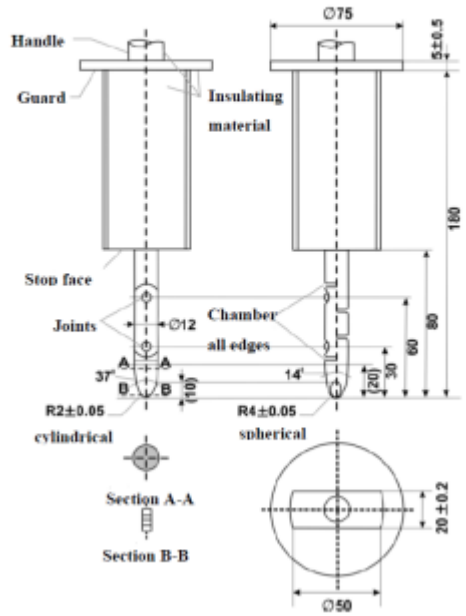
Internal electrical protection barriers are considered part of the enclosure

If appropriate a low-voltage supply (of not less than 40 V and not more than 50 V) in series with a suitable lamp should be connected, between the jointed test finger and high voltage live parts inside the electrical protection barrier or enclosure.

Figure 3 **Jointed Test Finger**

Access probe
(Dimensions in mm)

IPXXB Jointed text finger



Material: metal, except where otherwise specified

Linear dimensions in mm.

Tolerances on dimensions without specific tolerance:

(a) on angles: +0/-10 seconds;

(b) on linear dimensions:

(i) up to 25 mm: +0/-0.05;

(ii) over 25 mm: ±0.2.

Both joints shall permit movement in the same plane and the same direction through an angle of 90° with a 0 to +10° tolerance.

The requirements of paragraph 5.2.8.1.3. of this Regulation are met if the jointed test finger described in Figure 3, is unable to contact high voltage live parts.

If necessary a mirror or a fiberscope may be used in order to inspect whether the jointed test finger touches the high voltage buses.

If this requirement is verified by a signal circuit between the jointed test finger and high voltage live parts, the lamp shall not light.

4.1. Test method for measuring electric resistance:

(a) Test method using a resistance tester.

The resistance tester is connected to the measuring points (typically, electrical chassis and electro conductive enclosure/electrical protection barrier) and the resistance is measured using a resistance tester that meets the specification that follows:

(i) Resistance tester: Measurement current at least 0.2 A;

(ii) Resolution: 0.01 Ω or less;

(iii) The resistance R shall be less than 0.1 Ω.

(b) Test method using DC power supply, voltmeter and ammeter.

The DC power supply, voltmeter and ammeter are connected to the measuring points (Typically, electrical chassis and electro conductive enclosure/electrical protection barrier).

The voltage of the DC power supply is adjusted so that the current flow becomes at least 0.2 A.

The current "I" and the voltage "U" are measured.

The resistance "R" is calculated according to the following formula:

$R = U / I$

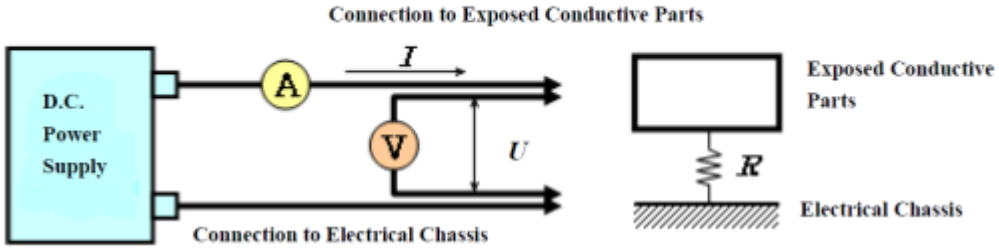
The resistance R shall be less than 0.1 Ω.

Note: If lead wires are used for voltage and current measurement, each lead wire shall be independently connected to the electrical protection barrier/enclosure/electrical chassis. Terminal can be common for voltage measurement and current measurement.

Example of the test method using DC power supply, voltmeter and ammeter is shown below.

Figure 4 **Example of theTest Method using DC Power Supply**

Figure 4
Example of theTest Method using DC Power Supply



5. Isolation resistance

5.1. General

The isolation resistance for each high voltage bus of the vehicle is measured or shall be determined by calculating the measurement values of each part or component unit of a high voltage bus.

All measurements for calculating voltage(s) and electrical isolation are made after a minimum of 10 seconds after the impact.

5.2. Measurement method

The isolation resistance measurement is conducted by selecting an appropriate measurement method from among those listed in paragraphs 5.2.1. to 5.2.2. of this Annex, depending on the electrical charge of the live parts or the isolation resistance.

The range of the electrical circuit to be measured is clarified in advance, using electrical circuit diagrams. If the high voltage buses are conductively isolated from each other, isolation resistance shall be measured for each electrical circuit.

Moreover, modifications necessary for measuring the isolation resistance may be carried out, such as removal of the cover in order to reach the live parts, drawing of measurement lines and change in software.

In cases where the measured values are not stable due to the operation of the on-board isolation resistance monitoring system, necessary modifications for conducting the measurement may be carried out by stopping the operation of the device concerned or by removing it. Furthermore, when the device is removed, a set of drawings will be used to prove that the isolation resistance between the live parts and the electrical chassis remains unchanged.

These modifications shall not influence the test results.

Utmost care shall be exercised to avoid short circuit and electric shock since this confirmation might require direct operations of the high-voltage circuit.

5.2.1. Measurement method using DC voltage from external sources

5.2.1.1. Measurement instrument

An isolation resistance test instrument capable of applying a DC voltage higher than the working voltage of the high voltage bus shall be used.

5.2.1.2. Measurement method

An isolation resistance test instrument is connected between the live parts and the electrical chassis. The isolation resistance is subsequently measured by applying a DC voltage at least half of the working voltage of the high voltage bus.

If the system has several voltage ranges (e.g. because of boost converter) in conductively connected circuit and some of the components cannot withstand the working voltage of the entire circuit, the isolation resistance between those components and the electrical chassis can be measured separately by applying at least half of their own working voltage with those components disconnected.

5.2.2. Measurement method using the vehicle's own REESS as DC voltage source.

5.2.2.1. Test vehicle conditions

The high voltage-bus is energized by the vehicle's own REESS and/or energy conversion system and the voltage level of the REESS and/or energy conversion system throughout the test shall be at least the nominal operating voltage as specified by the vehicle manufacturer.

5.2.2.2. Measurement instrument

The voltmeter used in this test shall measure DC values and have an internal resistance of at least 10 MΩ.

5.2.2.3. Measurement method

5.2.2.3.1. First step

The voltage is measured as shown in Figure 1 and the high voltage bus voltage (U_b) is recorded. U_b shall be equal to or greater than the nominal operating voltage of the REESS and/or energy conversion system as specified by the vehicle manufacturer.

5.2.2.3.2. Second step

The voltage (U_1) between the negative side of the high voltage bus and the electrical chassis is measured and recorded (see Figure 1).

5.2.2.3.3. Third step

The voltage (U_2) between the positive side of the high voltage bus and the electrical chassis is measured and recorded (see Figure 1).

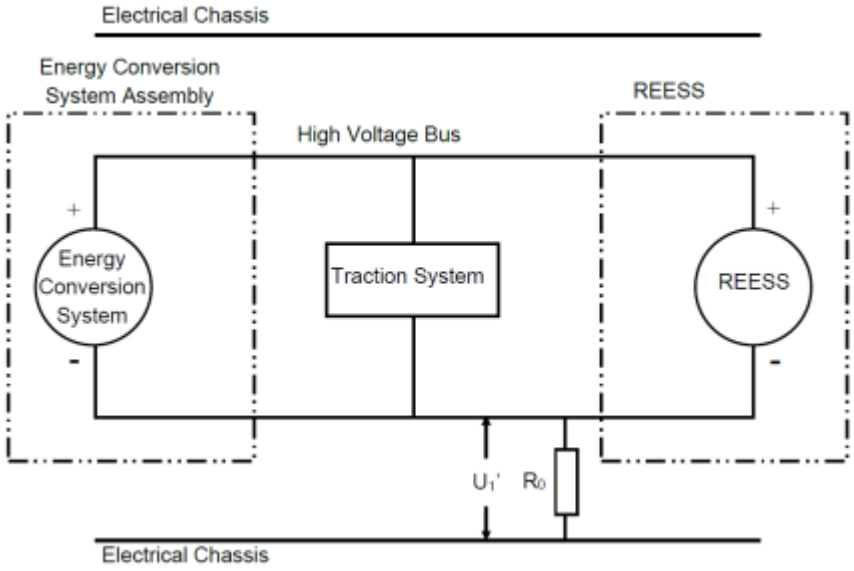
5.2.2.3.4. Fourth step

If U_1 is greater than or equal to U_2, a standard known resistance (Ro) is inserted between the negative side of the high voltage bus and the electrical chassis. With Ro installed, the voltage (U_1') between the negative side of the high voltage bus and the electrical chassis is measured (see Figure 5).

The electrical isolation (Ri) is calculated according to the following formula:

$Ri = Ro*U_b*(1/U_1' - 1/U_1)$

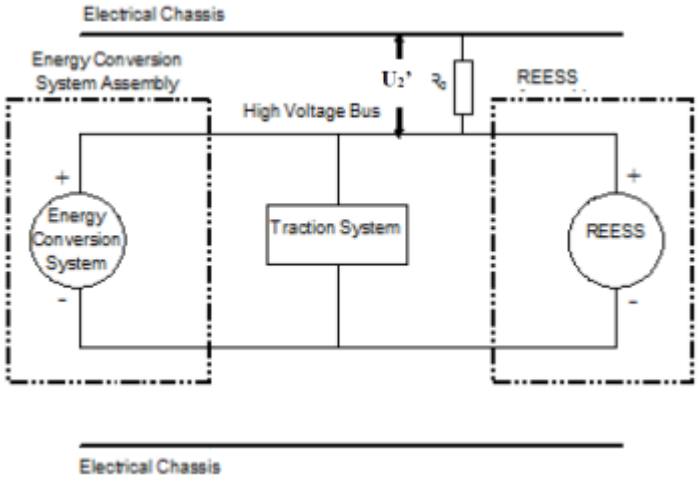
Figure 5 **Measurement of U_1'**



If U_2 is greater than U_1, insert a standard known resistance (R_o) between the positive side of the high voltage bus and the electrical chassis. With R_o installed, measure the voltage (U_2') between the positive side of the high voltage bus and the electrical chassis (see Figure 6 below). The electrical isolation (Ri) is calculated according to the following formula:

$R_i = R_o * U_b * (1/U_2' - 1/U_2)$

Figure 6 **Measurement of U_2'**



5.2.2.3.5. Fifth step.

The electrical isolation value Ri (in Ω) divided by the working voltage of the high voltage bus (in V) results in the isolation resistance (in Ω/V).

Note: The standard known resistance R_o (in Ω) should be the value of the minimum required isolation resistance (Ω/V) multiplied by the working voltage (V) of the vehicle plus/minus 20 per cent. R_o is not required to be precisely this value since the equations are valid for any R_o; however, a R_o value in this range should provide a good resolution for the voltage measurements.

6. Electrolyte leakage

An appropriate coating, if necessary, may be applied to the physical protection (casing) in order to confirm if there is any electrolyte leakage from the REESS resulting from the test. Unless the manufacturer provides means to

differentiate between the leakage of different liquids, all liquid leakage shall be considered as the electrolyte.

7. REESS retention

Compliance shall be determined by visual inspection.

Annex 10 Test Conditions and Procedures for the Assessment of Post-Crash Hydrogen Fuel System Integrity

1. Purpose

Determination of compliance with the requirements of paragraph 5.2.7. of this Regulation.

2. Definitions

For the purposes of this Annex:

2.1. "*Enclosed spaces"* means the special volumes within the vehicle (or the vehicle outline across openings) that are external to the hydrogen system (storage system, fuel cell system, internal combustion engine (ICE) and fuel flow management system).

2.2. "*Luggage compartment"* means the space in the vehicle for luggage and/or goods accommodation, bounded by the roof, hood, floor, side walls, being separated from the passenger compartment by the front bulkhead or the rear bulkhead.

2.3. "*Nominal working pressure (NWP)"* means the gauge pressure that characterizes typical operation of a system. For compressed hydrogen gas containers, NWP is the settled pressure of compressed gas in a fully fuelled container or storage system at a uniform temperature of 15 °C.

3. Preparation, Instrumentation and Test Conditions

3.1. Compressed hydrogen storage systems and downstream piping

3.1.1. Prior to conducting the crash test, instrumentation is installed in the hydrogen storage system to perform the required pressure and temperature measurements if the standard vehicle does not already possess instrumentation with the required accuracy.

3.1.2. The hydrogen storage system is then purged, if necessary, following manufacturer directions to remove impurities from the container before filling the storage system with compressed hydrogen or helium gas. Since the storage system pressure varies with temperature, the targeted fill pressure is a function of the temperature. The target pressure shall be determined from the following equation:

$P_{target} = NWP \times (273 + T_o) / 288$

where NWP is the nominal working pressure (MPa), T_o is the ambient temperature to which the storage system is expected to settle, and P_{target} is the targeted fill pressure after the temperature settles.

3.1.3. The container is filled to a minimum of 95 per cent of the targeted fill pressure and allowed to settle (stabilize) prior to conducting the crash test.

3.1.4. The main stop valve and shut-off valves for hydrogen gas, located in the downstream hydrogen gas piping, are in the normal driving condition kept open immediately prior to the impact.

3.2. Enclosed spaces

3.2.1. Sensors are selected to measure either the build-up of the hydrogen or helium gas or the reduction in oxygen (due to displacement of air by leaking hydrogen/helium).

3.2.2. Sensors are calibrated to traceable references to ensure an accuracy of ±5 per cent at the targeted criteria of 4 per cent hydrogen or 3 per cent helium by volume in air, and a full scale measurement capability of at least 25

per cent above the target criteria. The sensor shall be capable of a 90 per cent response to a full scale change in concentration within 10 seconds.

3.2.3. Prior to the crash impact, the sensors are located in the passenger and luggage compartments of the vehicle as follows:

(a) At a distance within 250 mm of the headliner above the driver's seat or near the top centre of the passenger compartment;

(b) At a distance within 250 mm of the floor in front of the rear (or rear most) seat in the passenger compartment; and

(c) At a distance within 100 mm of the top of luggage compartments inside the vehicle that are not directly affected by the particular crash impact to be conducted.

3.2.4. The sensors are securely mounted on the vehicle structure or seats and protected for the planned crash test from debris, air bag exhaust gas and projectiles. The measurements following the crash are recorded by instruments located in the vehicle or by remote transmission.

3.2.5. The test may be conducted either outdoors in an area protected from the wind and possible solar effects, or indoors in a space that is large enough or ventilated to prevent the build-up of hydrogen to more than 10 per cent of the targeted criteria in the passenger and luggage compartments.

4. Post-crash Leak Test Measurement for a Compressed Hydrogen Storage System Filled with Compressed Hydrogen

4.1. The hydrogen gas pressure, P_0 (MPa), and temperature, T_0 (°C), are measured immediately before the impact and then at a time interval, Δt (min), after the impact.

4.1.1. The time interval, Δt, starts when the vehicle comes to rest after the impact and continues for at least 60 minutes.

4.1.2. The time interval, Δt shall be increased if necessary in order to accommodate measurement accuracy for a storage system with a large volume operating up to 70MPa; in that case, Δt can be calculated from the following formula:

$$\Delta t = V_{CHSS} \times NWP / 1{,}000 \times ((-0.027 \times NWP + 4) \times R_s - 0.21) - 1.7 \times R_s$$

where $R_s = P_s / NWP$, P_s is the pressure range of the pressure sensor (MPa), NWP is the Nominal Working Pressure (MPa), V_{CHSS} is the volume of the compressed hydrogen storage system (L), and Δt is the time interval (min).

4.1.3. If the calculated value of Δt is less than 60 minutes, Δt is set to 60 minutes.

4.2. The initial mass of hydrogen in the storage system can be calculated as follows:

$$P_o' = P_o \times 288 / (273 + T_0)$$

$$\rho_o' = -0.0027 \times (P_0')^2 + 0.75 \times P_0' + 1.07$$

$$M_o = \rho_o' \times V_{CHSS}$$

4.3. Correspondingly, the final mass of hydrogen in the storage system, M_f, at the end of the time interval, Δt, can be calculated as follows:

$$P_f' = P_f \times 288 / (273 + T_f)$$

$$\rho_f' = -0.0027 \times (P_f')^2 + 0.75 \times P_f' + 1.07$$

$$M_f = \rho_f' \times V_{CHSS}$$

where P_f is the measured final pressure (MPa) at the end of the time interval, and T_f is the measured final temperature (°C).

4.4. The average hydrogen flow rate over the time interval is therefore:

$V_{H2} = (M_f\text{-}M_o) / \Delta t \times 22.41 / 2.016 \times (P_{target} / P_o)$

where V_{H2} is the average volumetric flow rate (NL/min) over the time interval and the term (P_{target}/P_o) is used to compensate for differences between the measured initial pressure (P_o) and the targeted fill pressure (P_{target}).

5. Post-Crash Leak Test Measurement for a Compressed Hydrogen Storage System Filled with Compressed Helium

5.1. The helium gas pressure, P_0 (MPa), and temperature T_0 (°C), are measured immediately before the impact and then at a predetermined time interval after the impact.

5.1.1. The time interval, Δt, starts when the vehicle comes to rest after the impact and continues for at least 60 minutes.

5.1.2. The time interval, Δt, shall be increased, if necessary, in order to accommodate measurement accuracy for a storage system with a large volume operating up to 70 MPa; in that case, Δt can be calculated from the following equation:

$\Delta t = V_{CHSS} \times NWP / 1000 \times ((-0.028 \times NWP + 5.5) \times R_s - 0.3) - 2.6 \times R_s$

where $R_s = P_s / NWP$, P_s is the pressure range of the pressure sensor (MPa), NWP is the Nominal Working Pressure (MPa), V_{CHSS} is the volume of the compressed storage system (L), and Δt is the time interval (min).

5.1.3. If the value of Δt is less than 60 minutes, Δt is set to 60 minutes.

5.2. The initial mass of helium in the storage system is calculated as follows:

$P_o' = P_o \times 288 / (273 + T_0)$

$\rho_o' = -0.0043 \times (P_0')^2 + 1.53 \times P_0' + 1.49$

$M_o = \rho_o' \times V_{CHSS}$

5.3. The final mass of helium in the storage system at the end of the time interval, Δt, is calculated as follows:

$P_f' = P_f \times 288 / (273 + T_f)$

$\rho_f' = -0.0043 \times (P_f')^2 + 1.53 \times P_f' + 1.49$

$M_f = \rho_f' \times V_{CHSS}$

where P_f is the measured final pressure (MPa) at the end of the time interval, and T_f is the measured final temperature (°C).

5.4. The average helium flow rate over the time interval is therefore:

$V_{He} = (M_f\text{-}M_o) / \Delta t \times 22.41 / 4.003 \times (P_{target} / P_o)$

where V_{He} is the average volumetric flow rate (NL/min) over the time interval and the term (P_{target}/P_o) is used to compensate for differences between the measured initial pressure (P_o) and the targeted fill pressure (P_{target}).

5.5. Conversion of the average volumetric flow of helium to the average hydrogen flow is calculated with the following formula:

$V_{H2} = V_{He} / 0.75$

where V_{H2} is the corresponding average volumetric flow of hydrogen.

6. Post-Crash Concentration Measurement for Enclosed Spaces

6.1. Post-crash data collection in enclosed spaces commences when the vehicle comes to a rest. Data from the sensors installed in accordance with paragraph 3.2. of this annex are collected at least every five seconds and continue for a period of 60 minutes after the test. A first-order lag (time constant) up to a maximum of five seconds may be applied to the measurements to provide "smoothing" and filter the effects of spurious data points.

Specifications for the restraint system test (Full Width Rigid Barrier test)

In the case of a compressed hydrogen-fuelled vehicle, compliance with paragraphs 5.2.7.1. to 5.2.7.3. shall be shown.